

Coding Sample: Escaping Inequality in India

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What follows is the 138-page code appendix to my independent paper and writing sample, “Escaping Inequality in India: The Role of English-Medium Instruction.”

Due to the length of this code, the time-constrained reviewer is encouraged to prioritize these 47 pages, excerpted from the code chunks listed below them:

14-16. Scalable **import, cleaning, and merging** of multi-file census and geospatial data.

- Chunk 4: Import key data files

31-32. **QA checks, disambiguation of identifiers** for district-level crosswalk.

- Chunk 6: Match districts: Construct district tracking dfs

50-52. **Testing for systematic missingness** using the Benjamini–Hochberg procedure and parallelized logistic regressions.

- Chunk 8: Participation model: Are NAs randomly distributed

55-57. **Survey-weighted probit estimation**, average marginal effects derivation, and setup for sample selection correction under a complex survey design.

- Chunk 9: Participation model: Estimation and IMR
- Chunk 10: Participation model: Derive AME

80-88. Reusable **record-linkage functions** using cascading fuzzy matches and unmatched/false-match diagnostics to harmonize district identifiers across years and data sources.

- Chunk 16: Match districts: Test joining methods
- Chunk 17: Match districts: Define joining functions

105-107. Reusable **IV specification and estimation pipeline** with clustered inference, plus contiguity-based spatial weights and setup for spatially lagged models.

- Chunk 24: Geospatial: Neighbor list construction
- Chunk 25: 2SLS: Controls and 2SLS formula
- Chunk 26: 2SLS: Baseline 2SLS models

111-112. **Robustness checks and diagnostics** for multicollinearity, spatial autocorrelation.

- Chunk 28: Multicollinearity check
- Chunk 29: Geospatial: Spatial autocorrelation tests

117-138. **Automated generation of publication-quality figures and tables** (including both summary statistics and regression outputs).

- Chunks 31-40: [Output for paper]

Source: R/io/read_inputs.R

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```

path <- row$absolute_path[[1]]
file_type <- tolower(row$file_type[[1]])
switch(
  file_type,
  sav = read_sav_short(path),
  csv = read_csv_short(path),
  xls = read_excel_short(path),
  xlsx = read_excel_short(path),
  ods = read_ods_short(path),
  shp = {
    need_pkg("sf", "shapefiles")
    sf::st_read(path, quiet = TRUE)
  },
  shx = path,
  dbf = path,
  prj = path,
  png = path,
  stop("No reader implemented for ", file_type, call. = FALSE)
)
}

#' read all manifest rows for a source
#'
#' @return Named list of reader outputs keyed by manifest file_id.
read_manifest_group <- function(paths, source_id) {
  rows <- require_manifest_files(paths, source_id)
  stats::setNames(lapply(seq_len(nrow(rows)), function(i) read_by_manifest_row(rows[i, ])),
    ↪ rows$file_id)
}

```

QA and identifier-disambiguation checks

Source: R/districts/build_district_tracker.R

```

#' build district tracker
#'
build_district_tracker <- function(raw_district_changes) {
  tracker_key <- if ("india_district_tracker" %in% names(raw_district_changes))
    ↪ "india_district_tracker" else "tracker"
  parsed <- list(
    alluvial = parse_alluvial_district_changes(raw_district_changes[["alluvial"]] %||%
    ↪ data.frame()),
    carveouts_renamings =
    ↪ parse_carveouts_renamings(raw_district_changes[["carveouts_renamings"]] %||%
    ↪ data.frame()),
    new_districts_created =
    ↪ parse_new_districts_created(raw_district_changes[["new_districts_created"]] %||%
    ↪ data.frame()),
    name_changes = parse_name_changes(raw_district_changes[["name_changes"]] %||%
    ↪ data.frame()),
    district_splits = parse_district_splits(raw_district_changes[["district_splits"]] %||%
    ↪ data.frame())
  )
  parsed[[tracker_key]] <- parse_india_district_tracker(raw_district_changes[[tracker_key]]
  ↪ %||% data.frame())
}

```

```

    extras <- setdiff(names(raw_district_changes), c("alluvial", "india_district_tracker",
↪ "tracker", "carveouts_renamings", "new_districts_created", "name_changes",
↪ "district_splits"))
    for (nm in extras) parsed[[nm]] <-
      ↪ parse_district_change_source(raw_district_changes[[nm]], source_type = nm)
    standardize_tracker_names(standardize_tracker_years(combine_district_tracker_sources_
      ↪ (parsed)))
  }

#' parse alluvial district changes
#'
parse_alluvial_district_changes <- function(x) {
  parse_district_change_source(x, source_type = "alluvial")
}

#' parse india district tracker
#'
parse_india_district_tracker <- function(x) {
  parse_district_change_source(x, source_type = "india_district_tracker")
}

#' parse carveouts renamings
#'
parse_carveouts_renamings <- function(x) {
  parse_district_change_source(x, source_type = "carveouts_renamings")
}

#' parse new districts created
#'
parse_new_districts_created <- function(x) {
  parse_district_change_source(x, source_type = "new_districts_created")
}

#' parse name changes
#'
parse_name_changes <- function(x) {
  parse_district_change_source(x, source_type = "name_changes")
}

#' parse district splits
#'
parse_district_splits <- function(x) {
  parse_district_change_source(x, source_type = "district_splits")
}

parse_district_change_source <- function(x, source_type) {
  x <- safe_df(x)
  if (!nrow(x)) return(data.frame(source_type = character(), stringsAsFactors = FALSE))
  x$source_type <- source_type
  x$source_state_raw <- first_present_value(x, c("state", "State", "state_raw",
↪ "state_from", "state_01", "state_07", "state_08", "state_17", "state_18", "state_20"))
  x$source_district_raw <- first_present_value(x, c("district", "District", "district_raw",
↪ "district_from", "district_01", "district_07", "district_08", "district_17",
↪ "district_18", "district_20", "district_name"))

```

```

x$target_state_raw <- first_present_value(x, c("state_to", "new_state", "target_state",
↪ "state_20", "state_18", "state_17", "state_08", "state_07", "state"))
x$target_district_raw <- first_present_value(x, c("district_to", "new_district",
↪ "target_district", "district_20", "district_18", "district_17", "district_08",
↪ "district_07", "district"))
x$source_year_raw <- first_present_value(x, c("source_year", "year_from", "from_year",
↪ "start_year", "year"))
x$target_year_raw <- first_present_value(x, c("target_year", "year_to", "to_year",
↪ "end_year", "year"))
x$change_type <- first_present_value(x, c("change_type", "type", "event_type", "status"))
x
}

first_present_value <- function(df, candidates) {
  hit <- first_col(df, candidates)
  if (is.null(hit)) return(rep(NA_character_, nrow(df)))
  as.character(df[[hit]])
}

#' combine district tracker sources
#'
combine_district_tracker_sources <- function(raw_district_changes) {
  out <- safe_bind_rows(lapply(names(raw_district_changes), function(name) {
    x <- safe_df(raw_district_changes[[name]])
    if (!nrow(x)) return(data.frame())
    x$source_file_id <- if ("source_file_id" %in% names(x)) x$source_file_id else name
    if (!"source_type" %in% names(x)) x$source_type <- name
    x
  })))
  if (nrow(out)) out$.row_in_source <- ave(seq_len(nrow(out)), out$source_file_id, FUN =
    ↪ seq_along)
  out
}

#' standardize tracker years
#'
standardize_tracker_years <- function(tracker) {
  tracker <- safe_df(tracker)
  if (!nrow(tracker)) return(tracker)
  if ("source_year_raw" %in% names(tracker) && !"source_year" %in% names(tracker))
    ↪ tracker$source_year <- suppressWarnings(as.integer(tracker$source_year_raw))
  if ("target_year_raw" %in% names(tracker) && !"target_year" %in% names(tracker))
    ↪ tracker$target_year <- suppressWarnings(as.integer(tracker$target_year_raw))
  tracker
}

#' standardize tracker names
#'
standardize_tracker_names <- function(tracker) {
  tracker <- safe_df(tracker)
  if (!nrow(tracker)) return(tracker)
  if ("source_state_raw" %in% names(tracker)) tracker$source_state_key <-
    ↪ canonicalize_state_name(tracker$source_state_raw)
  if ("source_district_raw" %in% names(tracker)) tracker$source_district_key <-
    ↪ canon(tracker$source_district_raw)

```

```

if ("target_state_raw" %in% names(tracker)) tracker$target_state_key <-
  ↪ canonicalize_state_name(tracker$target_state_raw)
if ("target_district_raw" %in% names(tracker)) tracker$target_district_key <-
  ↪ canon(tracker$target_district_raw)
tracker
}

```

Parallelized missingness diagnostics with BH correction

Source: R/selection/diagnose_missingness.R

```

missingness_variables <- function(selection_data) {
  df <- as.data.frame(selection_data, stringsAsFactors = FALSE)
  all_child_missing_vars <- c("DIST_FROM_NEAREST_PRIMARY_CLASS",
  ↪ "dmean_num_ENROLLMENT_COST", "father_educ")
  enrolled_only_missing_vars <- c("TUTION_FEE", "EXAMINATION_FEE", "OTHER_FEES_PAYMENTS",
  ↪ "BOOKS", "STATIONERY", "UNIFORM", "TRANSPORT")

  # Legacy Chunk 8 distinguished variables used in the probit from
  # enrolled-only expenditure variables. Keep that distinction so the total
  # row labelled "probit-model" is not inflated by fee variables that are
  # undefined for non-enrolled children.
  probit_vars <- c(
    "enrolled", "ENROLLED", "AGE", "age", "SEX", "HH_SIZE", "RELIGION", "SOCIAL_GROUP",
    "SECTOR", "state_0708", "region_0708", all_child_missing_vars
  )
  probit_vars <- intersect(probit_vars, names(df))
  if (!length(probit_vars)) probit_vars <- setdiff(names(df),
  ↪ intersect(enrolled_only_missing_vars, names(df)))

  list(
    probit_vars = probit_vars,
    miss_vars_all = intersect(all_child_missing_vars, names(df)),
    miss_vars_enrolled = intersect(enrolled_only_missing_vars, names(df)),
    group_vars = intersect(c("SECTOR", "SEX", "RELIGION", "SOCIAL_GROUP", "state_0708"),
    ↪ names(df)),
    cts_vars = intersect(c("AGE", "HH_SIZE"), names(df)),
    regional_vars = intersect(c("state_0708", "region_0708"), names(df))
  )
}

#' diagnose missingness
#'
#' Port the legacy Chunk 8 missingness diagnostics into an opt-in diagnostic
#' object. The returned list preserves the legacy sequence: variable counts,
#' regional rankings, missingness correlation matrices, BH-adjusted logit screens,
#' and notes for commented case-study / chi-square checks.
diagnose_missingness <- function(selection_data, cfg) {
  df <- as.data.frame(selection_data, stringsAsFactors = FALSE)
  vars <- missingness_variables(df)
  probit_vars <- vars$probit_vars

  counts <- summarize_missingness_by_variable(df[probit_vars])
  if (length(probit_vars)) {
    total_na <- sum(!stats::complete.cases(df[probit_vars]))
  }
}

```

```

    total_complete <- sum(stats::complete.cases(df[probit_vars]))
  } else {
    total_na <- 0L
    total_complete <- nrow(df)
  }
counts <- safe_bind_rows(list(
  counts,
  data.frame(missing_var = "Total probit-model with NA", n_missing = total_na, pct_missing
    ↪ = if (nrow(df)) total_na / nrow(df) else NA_real_),
  data.frame(missing_var = "Total probit-model with no NA", n_missing = total_complete,
    ↪ pct_missing = if (nrow(df)) total_complete / nrow(df) else NA_real_)
))

regional <- summarize_missingness_regions(df, probit_vars, vars)
corr_all <- missingness_correlation_matrix(df, vars$miss_vars_all, vars$group_vars,
  ↪ vars$cts_vars)
enrolled_rows <- enrolled_schooling_rows(df)
corr_enrolled <- missingness_correlation_matrix(df[enrolled_rows, , drop = FALSE],
  ↪ vars$miss_vars_enrolled, vars$group_vars, vars$cts_vars)

covars <- intersect(c("SECTOR", "SEX", "AGE", "HH_SIZE", "RELIGION", "SOCIAL_GROUP",
  ↪ "state_0708"), names(df))
miss_all <- intersect(vars$miss_vars_all, names(df))
miss_enrolled <- intersect(vars$miss_vars_enrolled, names(df))
logit_all <- if (length(miss_all) && length(covars)) run_missingness_logits(df, miss_all,
  ↪ covars) else data.frame()
logit_enrolled <- if (length(miss_enrolled) && length(covars) && any(enrolled_rows))
  ↪ run_missingness_logits(df[enrolled_rows, , drop = FALSE], miss_enrolled, covars) else
  ↪ data.frame()
logit_summary <- summarize_missingness_logits(safe_bind_rows(list(logit_all,
  ↪ logit_enrolled)))
case_study <- summarize_missingness_case_study(df, vars)
chi_square <- summarize_missingness_chisq(df, probit_vars)

notes <- data.frame(
  diagnostic = c(
    "rajasthan_southern_case_study",
    "chi_square_any_na_by_state",
    "chi_square_any_na_by_region",
    "parallel_missingness_logit"
  ),
  legacy_status = c(
    "commented diagnostic View() code preserved as documented note",
    "commented diagnostic test not run by default",
    "commented diagnostic test not run by default",
    "ported using lapply for reproducible targets execution; legacy mclapply/parLapply
    ↪ choice documented"
  ),
  stringsAsFactors = FALSE
)

out <- list(
  missing_counts = counts,
  regional_cost = regional$cost,
  regional_distance = regional$distance,

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    regional_father_education = regional$father_education,
    corr_all = corr_all,
    corr_enrolled = corr_enrolled,
    logit_all = logit_all,
    logit_enrolled = logit_enrolled,
    logit_summary = logit_summary,
    case_study = case_study,
    chi_square = chi_square,
    notes = notes
  )
  class(out) <- c("emi_missingness_diagnostics", class(out))
  out
}

enrolled_schooling_rows <- function(df) {
  if (!"enrolled" %in% names(df)) return(rep(TRUE, nrow(df)))
  value <- tolower(as.character(df$enrolled))
  value %in% c("yes", "1", "true", "enrolled")
}

summarize_missingness_by_variable <- function(df) {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  if (!length(names(df))) {
    return(data.frame(missing_var = character(), n_missing = integer(), pct_missing =
      ↪ numeric(), stringsAsFactors = FALSE))
  }
  data.frame(
    missing_var = names(df),
    n_missing = vapply(df, function(x) sum(is.na(x)), integer(1)),
    pct_missing = if (nrow(df)) vapply(df, function(x) mean(is.na(x)), numeric(1)) else
      ↪ NA_real_,
    stringsAsFactors = FALSE
  )
}

summarize_missingness_case_study <- function(df, vars) {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  if (!nrow(df)) return(data.frame())
  state_col <- intersect(c("state_0708", "state"), names(df))
  region_col <- intersect(c("region_0708", "region"), names(df))
  if (!length(state_col)) return(data.frame(note = "No state column available for
    ↪ Rajasthan/Southern case study.", stringsAsFactors = FALSE))

  keep <- grepl("rajasthan", as.character(df[[state_col[[1]]]]), ignore.case = TRUE)
  case_scope <- "Rajasthan"
  if (length(region_col)) {
    region_keep <- grepl("southern", as.character(df[[region_col[[1]]]]), ignore.case =
      ↪ TRUE)
    if (any(keep & region_keep, na.rm = TRUE)) {
      keep <- keep & region_keep
      case_scope <- "Rajasthan / Southern"
    }
  }
}

vars_to_check <- intersect(c(vars$miss_vars_all, vars$miss_vars_enrolled), names(df))

```



```

if (!length(vars_to_check) || !any(keep, na.rm = TRUE)) {
  return(data.frame(case_scope = case_scope, n_rows = sum(keep, na.rm = TRUE), note = "No
    ↪ matching rows or missingness variables available.", stringsAsFactors = FALSE))
}
case_df <- df[keep, vars_to_check, drop = FALSE]
any_missing <- !stats::complete.cases(case_df)
data.frame(
  case_scope = case_scope,
  n_rows = nrow(case_df),
  n_rows_with_any_missing = sum(any_missing),
  n_rows_with_partial_missing = sum(any_missing & rowSums(is.na(case_df)) <
    ↪ ncol(case_df)),
  n_missing_cells = sum(is.na(case_df)),
  n_observed_cells_in_rows_with_missing = sum(!is.na(case_df[any_missing, , drop =
    ↪ FALSE])),
  interpretation = "Current analog of the legacy Rajasthan/Southern View() check: rows
    ↪ with partial cost-variable missingness preserve the legacy concern that a child was
    ↪ not necessarily excluded wholesale when one cost field was missing.",
  stringsAsFactors = FALSE
)
}

summarize_missingness_chisq <- function(df, probit_vars) {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  if (!nrow(df) || !length(probit_vars)) return(data.frame())
  probit_vars <- intersect(probit_vars, names(df))
  if (!length(probit_vars)) return(data.frame())
  any_na <- !stats::complete.cases(df[probit_vars])
  safe_test <- function(group_col, label) {
    if (!group_col %in% names(df)) {
      return(data.frame(test = label, status = "not_available", reason = paste("Missing
        ↪ column", group_col), stringsAsFactors = FALSE))
    }
    group <- as.character(df[[group_col]])
    keep <- !is.na(group) & nzchar(group)
    if (length(unique(group[keep])) < 2L || length(unique(any_na[keep])) < 2L) {
      return(data.frame(test = label, status = "not_estimated", reason = "Insufficient
        ↪ variation for chi-square test.", stringsAsFactors = FALSE))
    }
    tab <- table(any_na = any_na[keep], group = group[keep])
    fit <- suppressWarnings(stats::chisq.test(tab))
    data.frame(
      test = label,
      status = "estimated",
      statistic = unname(fit$statistic),
      parameter = unname(fit$parameter),
      p.value = unname(fit$p.value),
      n = sum(tab),
      method = fit$method,
      stringsAsFactors = FALSE
    )
  }
}
safe_bind_rows(list(
  safe_test("state_0708", "any_probit_model_na_by_state"),
  safe_test("region_0708", "any_probit_model_na_by_region")

```

```

))
}

summarize_missingness_regions <- function(df, probit_vars, vars, top_n = 20L) {
  empty <- data.frame()
  if (!"state_0708" %in% names(df)) {
    return(list(cost = empty, distance = empty, father_education = empty))
  }
  if (!length(probit_vars)) probit_vars <- names(df)
  temp <- df
  region_cols <- intersect(c("state_0708", "region_0708"), names(temp))
  # Legacy Chunk 8 inspected both state and region. Some cleaned selection
  # inputs no longer expose region_0708, so keep the diagnostic alive at the
  # state level instead of writing empty CSVs.
  if (!"region_0708" %in% region_cols) {
    temp$region_0708 <- "all_regions_available_in_state_only_input"
    region_cols <- c("state_0708", "region_0708")
  }

  temp$any_na_row <- !stats::complete.cases(temp[intersect(probit_vars, names(temp))])
  for (nm in c("dmean_num_ENROLLMENT_COST", "DIST_FROM_NEAREST_PRIMARY_CLASS",
    ↪ "father_educ")) {
    temp[[paste0("miss_", nm)]] <- if (nm %in% names(temp)) as.integer(is.na(temp[[nm]]))
    ↪ else NA_integer_
  }
  temp$is_urban <- if ("SECTOR" %in% names(temp)) as.integer(temp$SECTOR == "Urban") else
  ↪ NA_integer_
  temp$is_female <- if ("SEX" %in% names(temp)) as.integer(temp$SEX == "Female") else
  ↪ NA_integer_
  temp$is_hindu <- if ("RELIGION" %in% names(temp)) as.integer(temp$RELIGION == "Hindu")
  ↪ else NA_integer_
  temp$is_muslim <- if ("RELIGION" %in% names(temp)) as.integer(temp$RELIGION == "Muslim")
  ↪ else NA_integer_
  temp$is_st_sc_obc <- if ("SOCIAL_GROUP" %in% names(temp)) as.integer(temp$SOCIAL_GROUP
  ↪ %in% c("Scheduled Tribe", "Scheduled Caste", "Other Backward Class")) else NA_integer_

  measure_cols <- c("any_na_row", "miss_dmean_num_ENROLLMENT_COST",
  ↪ "miss_DIST_FROM_NEAREST_PRIMARY_CLASS", "miss_father_educ", "is_urban", "is_female",
  ↪ "is_hindu", "is_muslim", "is_st_sc_obc")
  grouped <- stats::aggregate(
    temp[measure_cols],
    temp[region_cols],
    function(x) mean(as.numeric(x), na.rm = TRUE)
  )
  n <- stats::aggregate(temp$any_na_row, temp[region_cols], length)
  names(n)[names(n) == "x"] <- "n"
  out <- merge(n, grouped, by = region_cols, all = TRUE)
  names(out) <- sub("^any_na_row$", "pct_any_na", names(out))
  out$region_diagnostic_level <- if ("region_0708" %in% names(df)) "state_region" else
  ↪ "state_only_fallback"
  pct_cols <- setdiff(names(out), c(region_cols, "n", "region_diagnostic_level"))
  out[pct_cols] <- lapply(out[pct_cols], function(x) round(100 * x, 2))
  rank_one <- function(col) {
    if (!col %in% names(out)) return(empty)
    out[order(-out[[col]]), , drop = FALSE][seq_len(min(top_n, nrow(out))), , drop = FALSE]
  }

```

```

}
list(
  cost = rank_one("miss_dmean_num_ENROLLMENT_COST"),
  distance = rank_one("miss_DIST_FROM_NEAREST_PRIMARY_CLASS"),
  father_education = rank_one("miss_father_educ")
)
}

missingness_correlation_matrix <- function(df, miss_vars, group_vars, cts_vars) {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  miss_vars <- intersect(miss_vars, names(df))
  if (!length(miss_vars) || !nrow(df)) return(matrix(numeric(), nrow = 0L, ncol = 0L))
  miss_df <- as.data.frame(lapply(df[miss_vars], function(x) as.integer(is.na(x))),
    ↪ check.names = FALSE)
  names(miss_df) <- paste0("miss_", names(miss_df))
  cts <- intersect(cts_vars, names(df))
  cts_df <- if (length(cts)) as.data.frame(lapply(df[cts], numeric_like), check.names =
    ↪ FALSE) else data.frame()
  grp <- intersect(group_vars, names(df))
  grp_df <- if (length(grp)) {
    grp_data <- as.data.frame(lapply(df[grp], function(x) {
      x <- as.character(x)
      x[is.na(x) | !nzchar(x)] <- "missing"
      factor(x)
    }), check.names = FALSE)
    # Legacy Chunk 8 used model.matrix-style expansions of sector, sex,
    # religion, social group, state, and related grouping variables before
    # computing missingness correlations. Small diagnostic subsets, especially
    # enrolled-only slices used in tests or regional filters, may contain only a
    # single observed level for one or more grouping variables. model.matrix()
    # cannot apply contrasts to one-level factors, so drop constant grouping
    # variables while preserving all grouping variables with real variation.
    varying <- vapply(grp_data, function(x) length(unique(as.character(x))) > 1L,
    ↪ logical(1))
    grp_data <- grp_data[varying]
    if (length(grp_data)) {
      stats::model.matrix(~ . - 1, data = grp_data)
    } else {
      matrix(numeric(), nrow = nrow(df), ncol = 0L)
    }
  } else matrix(numeric(), nrow = nrow(df), ncol = 0L)
  safe_pairwise_cor(cbind(miss_df, cts_df, grp_df))
}

#' check missing logit parallel
#'
#' Legacy Chunk 8 used mclapply/parLapply after defining the same one-logit-per-
#' missing-variable problem. Targets already parallelizes at the target level,
#' so this implementation keeps deterministic in-process lapply while preserving
#' the model specification and BH adjustment.
check_missing_logit_parallel <- function(df, miss_vars, covars, method_p = "BH") {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  miss_vars <- intersect(miss_vars, names(df))
  covars <- intersect(covars, names(df))
  if (!length(miss_vars) || !length(covars)) return(data.frame())

```

```

rhs <- paste(covars, collapse = " + ")
fit_one <- function(m) {
  y <- is.na(df[[m]])
  if (length(unique(y)) < 2L) return(data.frame())
  f <- stats::as.formula(paste0("is.na(`", m, "`) ~ ", rhs))
  tryCatch({
    fit <- stats::glm(f, data = df, family = stats::binomial)
    pseudoR2 <- 1 - fit$deviance / fit$null.deviance
    out <- broom::tidy(fit)
    out$missing_var <- m
    out$pseudoR2 <- pseudoR2
    out$noobs <- stats::noobs(fit)
    out
  }, error = function(e) {
    data.frame(term = NA_character_, estimate = NA_real_, std.error = NA_real_, statistic
      ↪ = NA_real_, p.value = NA_real_, missing_var = m, pseudoR2 = NA_real_, noobs =
      ↪ length(y), status = "failed", reason = conditionMessage(e), stringsAsFactors =
      ↪ FALSE)
  })
}
out <- safe_bind_rows(lapply(miss_vars, fit_one))
adjust_missingness_pvalues_bh(out, method_p = method_p)
}

run_missingness_logits <- function(df, miss_vars, covars) {
  check_missing_logit_parallel(df, miss_vars, covars)
}

adjust_missingness_pvalues_bh <- function(df, method_p = "BH") {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  if (!nrow(df) || !"p.value" %in% names(df)) return(df)
  df$p_adj <- NA_real_
  idx <- !is.na(df$p.value) & df$term != "(Intercept)"
  df$p_adj[idx] <- stats::p.adjust(df$p.value[idx], method = method_p)
  df
}

summarize_missingness_logits <- function(df) {
  df <- as.data.frame(df, stringsAsFactors = FALSE)
  if (!nrow(df) || !all(c("missing_var", "p_adj", "pseudoR2") %in% names(df))) {
    return(data.frame(missing_var = character(), n_sig = integer(), pseudoR2 = numeric(),
      ↪ stringsAsFactors = FALSE))
  }
  split_df <- split(df, df$missing_var)
  out <- safe_bind_rows(lapply(split_df, function(x) {
    pmax <- suppressWarnings(max(x$pseudoR2, na.rm = TRUE))
    if (!is.finite(pmax)) pmax <- NA_real_
    data.frame(
      missing_var = x$missing_var[[1]],
      n_sig = sum(x$p_adj < 0.05 & x$term != "(Intercept)", na.rm = TRUE),
      pseudoR2 = pmax,
      stringsAsFactors = FALSE
    )
  })))
out[order(-out$pseudoR2), , drop = FALSE]

```

```

}

missingness_correlation_pairs <- function(mat, top_n = 50L) {
  mat <- as.matrix(mat)
  if (!length(mat) || nrow(mat) < 2L || ncol(mat) < 2L) {
    return(data.frame(var1 = character(), var2 = character(), correlation = numeric(),
      ↪ abs_correlation = numeric(), stringsAsFactors = FALSE))
  }
  mat[lower.tri(mat, diag = TRUE)] <- NA_real_
  idx <- which(is.finite(mat), arr.ind = TRUE)
  if (!nrow(idx)) {
    return(data.frame(var1 = character(), var2 = character(), correlation = numeric(),
      ↪ abs_correlation = numeric(), stringsAsFactors = FALSE))
  }
  out <- data.frame(
    var1 = rownames(mat)[idx[, "row"]],
    var2 = colnames(mat)[idx[, "col"]],
    correlation = mat[idx],
    stringsAsFactors = FALSE
  )
  out$abs_correlation <- abs(out$correlation)
  out <- out[order(-out$abs_correlation), , drop = FALSE]
  utils::head(out, top_n)
}

save_missingness_correlation_heatmap <- function(mat, path, max_vars = 35L, title =
  ↪ "Missingness correlation matrix") {
  mat <- as.matrix(mat)
  dir.create(dirname(path), recursive = TRUE, showWarnings = FALSE)
  if (!length(mat) || nrow(mat) < 2L || ncol(mat) < 2L) {
    grDevices::png(path, width = 900, height = 500, res = 120)
    graphics::plot.new()
    graphics::text(0.5, 0.5, "No correlation matrix available")
    grDevices::dev.off()
    return(normalizePath(path, mustWork = FALSE))
  }
  score <- apply(abs(mat), 1L, function(x) max(x[is.finite(x) & x < 1], na.rm = TRUE))
  score[!is.finite(score)] <- 0
  keep <- names(sort(score, decreasing = TRUE))[seq_len(min(max_vars, length(score)))]
  mat <- mat[keep, keep, drop = FALSE]
  grDevices::png(path, width = 1400, height = 1200, res = 140)
  old <- graphics::par(no.readonly = TRUE)
  on.exit({ graphics::par(old); grDevices::dev.off() }, add = TRUE)
  graphics::par(mar = c(11, 11, 4, 2))
  graphics::image(
    x = seq_len(ncol(mat)),
    y = seq_len(nrow(mat)),
    z = t(mat[nrow(mat):1, , drop = FALSE]),
    axes = FALSE,
    xlab = "",
    ylab = "",
    main = title,
    zlim = c(-1, 1)
  )
}

```

```

graphics::axis(1, at = seq_len(ncol(mat)), labels = colnames(mat), las = 2, cex.axis =
↪ 0.55)
graphics::axis(2, at = seq_len(nrow(mat)), labels = rev(rownames(mat)), las = 2, cex.axis
↪ = 0.55)
graphics::box()
normalizePath(path, mustWork = FALSE)
}

```

```

save_missingness_logit_plot <- function(summary_df, path, title = "Structured missingness
↪ logit screen", top_n = 25L) {
  dir.create(dirname(path), recursive = TRUE, showWarnings = FALSE)
  df <- as.data.frame(summary_df, stringsAsFactors = FALSE)
  grDevices::png(path, width = 1100, height = 850, res = 130)
  old <- graphics::par(no.readonly = TRUE)
  on.exit({ graphics::par(old); grDevices::dev.off() }, add = TRUE)
  if (!nrow(df) || !all(c("missing_var", "pseudoR2") %in% names(df))) {
    graphics::plot.new()
    graphics::text(0.5, 0.5, "No missingness-logit summary available")
    return(normalizePath(path, mustWork = FALSE))
  }
  df <- df[is.finite(df$pseudoR2), , drop = FALSE]
  if (!nrow(df)) {
    graphics::plot.new()
    graphics::text(0.5, 0.5, "No finite pseudo-R2 values available")
    return(normalizePath(path, mustWork = FALSE))
  }
  df <- df[order(df$pseudoR2, decreasing = TRUE), , drop = FALSE]
  df <- utils::head(df, top_n)
  graphics::par(mar = c(5, 13, 4, 2))
  graphics::barplot(
    rev(df$pseudoR2),
    names.arg = rev(df$missing_var),
    horiz = TRUE,
    las = 1,
    xlab = "Pseudo-R2 (predictability of missingness)",
    main = title
  )
  normalizePath(path, mustWork = FALSE)
}

```

```

save_missingness_diagnostics <- function(diagnostics, dir =
↪ "outputs/diagnostics/extended/missingness") {
  dir.create(dir, recursive = TRUE, showWarnings = FALSE)
  if (!inherits(diagnostics, "emi_missingness_diagnostics")) diagnostics <-
↪ list(missing_counts = as.data.frame(diagnostics))
  paths <- c(
    missing_counts = write_diagnostic_csv(diagnostics$missing_counts %||% data.frame(),
↪ file.path(dir, "missingness_counts.csv")),
    regional_cost = write_diagnostic_csv(diagnostics$regional_cost %||% data.frame(),
↪ file.path(dir, "regional_missingness_cost.csv")),
    regional_distance = write_diagnostic_csv(diagnostics$regional_distance %||%
↪ data.frame(), file.path(dir, "regional_missingness_distance.csv")),
    regional_father = write_diagnostic_csv(diagnostics$regional_father_education %||%
↪ data.frame(), file.path(dir, "regional_missingness_father_education.csv")),
  )
}

```

```

logit_all = write_diagnostic_csv(diagnostics$logit_all %||% data.frame(), file.path(dir,
  ↪ "missingness_logits_all.csv")),
logit_enrolled = write_diagnostic_csv(diagnostics$logit_enrolled %||% data.frame(),
  ↪ file.path(dir, "missingness_logits_enrolled.csv")),
logit_summary = write_diagnostic_csv(diagnostics$logit_summary %||% data.frame(),
  ↪ file.path(dir, "missingness_logit_summary.csv")),
case_study = write_diagnostic_csv(diagnostics$case_study %||% data.frame(),
  ↪ file.path(dir, "missingness_rajasthan_southern_case_study.csv")),
chi_square = write_diagnostic_csv(diagnostics$chi_square %||% data.frame(),
  ↪ file.path(dir, "missingness_chi_square_tests.csv")),
notes = write_diagnostic_csv(diagnostics$notes %||% data.frame(), file.path(dir,
  ↪ "missingness_legacy_notes.csv"))
)
if (length(diagnostics$corr_all)) {
  paths <- c(
    paths,
    corr_all = write_diagnostic_matrix(diagnostics$corr_all, file.path(dir,
      ↪ "missingness_correlation_all.csv")),
    corr_all_pairs =
      ↪ write_diagnostic_csv(missingness_correlation_pairs(diagnostics$corr_all),
      ↪ file.path(dir, "missingness_correlation_all_top_pairs.csv")),
    corr_all_heatmap = save_missingness_correlation_heatmap(diagnostics$corr_all,
      ↪ file.path(dir, "missingness_correlation_all.png"), title = "Missingness
      ↪ correlations: all observations")
  )
}
if (length(diagnostics$corr_enrolled)) {
  paths <- c(
    paths,
    corr_enrolled = write_diagnostic_matrix(diagnostics$corr_enrolled, file.path(dir,
      ↪ "missingness_correlation_enrolled.csv")),
    corr_enrolled_pairs =
      ↪ write_diagnostic_csv(missingness_correlation_pairs(diagnostics$corr_enrolled),
      ↪ file.path(dir, "missingness_correlation_enrolled_top_pairs.csv")),
    corr_enrolled_heatmap =
      ↪ save_missingness_correlation_heatmap(diagnostics$corr_enrolled, file.path(dir,
      ↪ "missingness_correlation_enrolled.png"), title = "Missingness correlations:
      ↪ enrolled observations")
  )
}
if (nrow(diagnostics$logit_summary %||% data.frame())) {
  paths <- c(
    paths,
    logit_pseudo_r2_plot = save_missingness_logit_plot(
      diagnostics$logit_summary,
      file.path(dir, "missingness_logit_pseudo_r2.png"),
      title = "How structured is missingness?"
    )
  )
}
output_manifest(paths)
}

```


Survey-weighted probit and IMR setup

Source: R/selection/estimate_selection_probit.R

```
selection_probit_variables <- function(selection_data) {
  controls <- c("AGE", "age", "SEX", "HH_SIZE", "RELIGION", "SOCIAL_GROUP", "SECTOR")
  exclusion_all_kids <- c("DIST_FROM_NEAREST_PRIMARY_CLASS", "father_educ")
  exclusion_district <- c(
    "dmean_num_IS_EDU_FREE", "dmean_num_TUTION_FEE_WAIVED",
    "dmean_num_RECD_SCHOLARSHIP_STIPEND", "dmean_num_RECD_TXT_BOOKS",
    "dmean_num_RECD_STATIONERY", "dmean_num_MID_DAY_MEAL_ETC_RECD",
    "dmean_num_ENROLLMENT_COST"
  )
  intersect(c(controls, exclusion_all_kids, exclusion_district), names(selection_data))
}

#' estimate selection probit
#'
estimate_selection_probit <- function(selection_data, cfg) {
  if (!"enrolled" %in% names(selection_data) || all(is.na(selection_data$enrolled))) {
    return(list(status = "out_of_active_pipeline", reason = "No enrolled variable."))
  }
  covars <- selection_probit_variables(selection_data)
  if (!length(covars)) {
    return(list(status = "out_of_active_pipeline", reason = "No probit covariates."))
  }
  f_probit <- stats::reformulate(covars, response = "enrolled")
  if (identical(cfg$mode, "final") && requireNamespace("survey", quietly = TRUE)) {
    design <- build_survey_design_selection(selection_data)
    if (!is.null(design)) {
      out <- fit_selection_probit(design, f_probit)
      attr(out, "selection_probit_formula") <- f_probit
      return(out)
    }
  }
  out <- stats::glm(f_probit, data = selection_data, family = stats::binomial(link =
    ↪ "probit"))
  attr(out, "selection_probit_formula") <- f_probit
  out
}

#' build survey design selection
#'
build_survey_design_selection <- function(selection_df) {
  psu <- first_col(selection_df, c("FSU_SL_NO", "fsu", "PSU", "psu"))
  weight <- first_col(selection_df, c("weight", "WEIGHT", "Multiplier", "multiplier"))
  strata_cols <- intersect(c("STATE", "STRATUM", "SUB_STRATUM_NO"), names(selection_df))
  if (is.null(psu) || is.null(weight) || !length(strata_cols)) return(NULL)
  options(survey.lonely.psu = "average")
  selection_df$.survey_strata <- interaction(selection_df[strata_cols], drop = TRUE)
  survey::svydesign(
    ids = stats::as.formula(paste0("~", psu)),
    strata = ~.survey_strata,
    weights = stats::as.formula(paste0("~", weight)),
    data = selection_df,
    nest = TRUE
  )
}
```



```

)
}

#' fit selection probit
#'
fit_selection_probit <- function(selection_design, f_probit) {
  survey::svyglm(f_probit, design = selection_design, family = stats::quasibinomial(link =
    ↪ "probit"))
}

#' compute inverse mills ratio
#'
compute_inverse_mills_ratio <- function(model, selection_df, f_probit = NULL) {
  if (is.null(f_probit)) f_probit <- attr(model, "selection_probit_formula") %||%
    ↪ stats::formula(model)
  needed <- all.vars(f_probit)
  needed <- intersect(needed, names(selection_df))
  comp_cases <- stats::complete.cases(selection_df[, needed, drop = FALSE])
  lp <- rep(NA_real_, nrow(selection_df))
  lp[comp_cases] <- stats::predict(model, newdata = selection_df[comp_cases, , drop =
    ↪ FALSE], type = "link")
  phi_lp <- stats::dnorm(lp)
  Phi_lp <- pmin(pmax(stats::pnorm(lp), .Machine$double.eps), 1 - .Machine$double.eps)
  enrolled <- as.character(selection_df$enrolled)
  selection_df$IMR <- ifelse(enrolled == "Yes", phi_lp / Phi_lp, ifelse(enrolled == "No",
    ↪ phi_lp / (1 - Phi_lp), NA_real_))
  selection_df
}

#' tidy selection model
#'
tidy_selection_model <- function(model) {
  broom::tidy(model)
}

```

Average marginal effects via automatic differentiation

Source: R/selection/compute_average_marginal_effects.R

```

# Use automatic differentiation for SE delta-method gradients, as in the legacy Rmd.

#' compute average marginal effects
#'
compute_average_marginal_effects <- function(selection_model, selection_data, cfg = list())
  ↪ {
  if (!inherits(selection_model, "glm")) {
    return(ame_out_of_pipeline(
      selection_model$status %||% "out_of_active_pipeline",
      selection_model$reason %||% "Selection model not estimated in smoke mode"
    ))
  }

  # `selection_model` is often a survey::svyglm object loaded from the targets
  # store. Survey lonely-PSU handling is an R option, not a durable model
  # attribute, so it may not be set in the downstream AME target even though it
  # was set when the model was estimated. Set it explicitly around all AME

```

```

# computations to make reruns deterministic and warning/error-free.
old_lonely_psu <- getOption("survey.lonely.psu")
old_domain_lonely <- getOption("survey.adjust.domain.lonely")
options(survey.lonely.psu = "average", survey.adjust.domain.lonely = TRUE)
on.exit({
  if (is.null(old_lonely_psu)) {
    options(survey.lonely.psu = NULL)
  } else {
    options(survey.lonely.psu = old_lonely_psu)
  }
  if (is.null(old_domain_lonely)) {
    options(survey.adjust.domain.lonely = NULL)
  } else {
    options(survey.adjust.domain.lonely = old_domain_lonely)
  }
}, add = TRUE)

#' run expression while muffling survey lonely-PSU warnings
#'
#' survey emits lonely-PSU warnings even when survey.lonely.psu is set to
#' average and the adjustment is intentional. Muffle only those handled
#' warnings inside the AME target so strict final builds remain warning-clean.
muffle_survey_lonely_psu_warnings <- function(expr) {
  withCallingHandlers(
    expr,
    warning = function(w) {
      msg <- conditionMessage(w)
      lonely_psu_warning <- grepl(
        "has only one PSU at stage",
        msg,
        fixed = TRUE
      )
      if (lonely_psu_warning) invokeRestart("muffleWarning")
    }
  )
}

if (!isTRUE(cfg$run_full_ame)) {
  return(format_ame_results(data.frame(
    term = names(stats::coef(selection_model)),
    estimate = unname(stats::coef(selection_model)),
    std.error = NA_real_,
    statistic = NA_real_,
    p.value = NA_real_,
    s.value = NA_real_,
    conf.low = NA_real_,
    conf.high = NA_real_,
    method = "coefficient_fallback",
    status = "estimated",
    reason = "Draft config uses coefficient fallback instead of full AME computation"
  )))
}

if (!requireNamespace("marginaleffects", quietly = TRUE)) {

```

```

    return(ame_out_of_pipeline(
      "out_of_active_pipeline",
      "Package marginalesffects is required for full AME computation"
    ))
  }

out <- muffle_survey_lonely_psu_warnings(
  tryCatch(
    compute_ames_autodiff(selection_model, selection_data),
    error = function(e) {
      fallback <- compute_ames_probit_analytic(selection_model, selection_data)
      fallback$method <- "delta_method_analytic_probit"
      fallback$reason <- NA_character_
      fallback
    }
  )
)
format_ame_results(out)
}

ame_out_of_pipeline <- function(status, reason) {
  data.frame(
    term = NA_character_,
    estimate = NA_real_,
    std.error = NA_real_,
    statistic = NA_real_,
    p.value = NA_real_,
    s.value = NA_real_,
    conf.low = NA_real_,
    conf.high = NA_real_,
    method = "not_run",
    status = status,
    reason = reason
  )
}

#' extract model-frame data and AME weights
#'
#' marginalesffects warns, correctly, when weighted/survey models are summarized
#' without an explicit `wts` argument. Build a newdata frame from the exact
#' model estimation sample and attach a synthetic weight column whenever the
#' fitted model exposes usable weights.
ame_model_weight <- function(model_data, model_weights = NULL) {
  weight_cols <- intersect(c("weight", "WEIGHT", "Multiplier", "multiplier", "(weights)"),
    names(model_data))
  if (length(weight_cols)) {
    return(suppressWarnings(as.numeric(model_data[[weight_cols[[1]]]])))
  }

  if (!is.null(model_weights) && length(model_weights) == nrow(model_data)) {
    return(suppressWarnings(as.numeric(model_weights)))
  }

  NULL
}

```

```

# build marginaleffects newdata and weights
#
# @return List with `data`, `wts`, and `has_explicit_weights`.
ame_model_data_and_weights <- function(model) {
  model_data <- as.data.frame(stats::model.frame(model))
  model_weights <- tryCatch(stats::weights(model), error = function(e) NULL)
  weight <- ame_model_weight(model_data, model_weights)

  if (!is.null(weight) && length(weight) == nrow(model_data) && any(is.finite(weight) &
    ↪ weight != 1)) {
    model_data$.ame_weight <- weight
    return(list(data = model_data, wts = ".ame_weight", has_explicit_weights = TRUE))
  }

  list(data = model_data, wts = FALSE, has_explicit_weights = FALSE)
}

# run marginaleffects while muffling only a redundant explicit-weight warning
#
run_avg_slopes <- function(model, model_data, wts) {
  withCallingHandlers(
    marginaleffects::avg_slopes(
      model,
      newdata = model_data,
      wts = wts,
      vcov = TRUE,
      type = "response"
    ),
    warning = function(w) {
      msg <- conditionMessage(w)
      explicit_weight_warning <- grepl(
        "normally good practice to specify weights using the `wts` argument",
        msg,
        fixed = TRUE
      )
      if (explicit_weight_warning && !identical(wts, FALSE)) {
        invokeRestart("muffleWarning")
      }
    }
  )
}

# compute ames autodiff
#
compute_ames_autodiff <- function(model, newdata) {
  # Use the exact estimation sample retained by glm/svyglm. Passing the full
  # pre-model selection_data can have extra rows omitted during model fitting,
  # which makes marginaleffects recycle weights/covariates silently.
  amed <- ame_model_data_and_weights(model)
  run_avg_slopes(model, amed$data, amed$wts)
}

# compute ames fast draft
#

```

```

compute_ames_fast_draft <- function(model, newdata, n = 200) {
  amed <- ame_model_data_and_weights(model)
  run_avg_slopes(
    model,
    dplyr::slice_sample(amed$data, n = min(n, nrow(amed$data))),
    amed$wts
  )
}

#' compute analytic probit AMEs
#'
#' @return Data frame of observed-data average marginal effects.
compute_ames_probit_analytic <- function(model, newdata) {
  coefs <- stats::coef(model)
  coef_table <- as.data.frame(summary(model)$coefficients)
  terms <- setdiff(names(coefs), "(Intercept)")
  if (!length(terms)) return(ame_out_of_pipeline("out_of_active_pipeline", "Selection model
  ↪ has no non-intercept terms.))

  newdata <- stats::model.frame(model)
  weights <- if ("weight" %in% names(newdata)) num(newdata$weight) else rep(1,
  ↪ nrow(newdata))
  eta <- stats::predict(model, type = "link")
  mean_phi <- wmean(stats::dnorm(eta), weights)
  factor_levels <- model$xlevels %||% list()

  rows <- lapply(terms, function(term) {
    factor_var <- names(factor_levels)[vapply(names(factor_levels), function(v)
    ↪ startsWith(term, v), logical(1))]
    estimate <- NA_real_
    if (length(factor_var)) {
      var <- factor_var[[which.max(nchar(factor_var))]]
      level <- sub(paste0("^", var), "", term)
      if (level %in% factor_levels[[var]]) {
        hi <- newdata
        lo <- newdata
        hi[[var]] <- factor(level, levels = factor_levels[[var]])
        lo[[var]] <- factor(factor_levels[[var]][[1]], levels = factor_levels[[var]])
        estimate <- wmean(stats::predict(model, newdata = hi, type = "response") -
    ↪ stats::predict(model, newdata = lo, type = "response"), weights)
      }
    } else {
      estimate <- unname(coefs[[term]]) * mean_phi
    }
    p_col <- intersect(c("Pr(>|z|)", "Pr(>|t|)"), names(coef_table))
    p <- if (term %in% rownames(coef_table) && length(p_col)) coef_table[term, p_col[[1]]]
    ↪ else NA_real_
    se_col <- intersect(c("Std. Error", "std.error"), names(coef_table))
    se_beta <- if (term %in% rownames(coef_table) && length(se_col)) coef_table[term,
    ↪ se_col[[1]]] else NA_real_
    se <- if (is.finite(se_beta)) abs(se_beta * mean_phi) else NA_real_
    z <- if (is.finite(se) && se > 0) estimate / se else NA_real_
    p_out <- if (is.finite(z)) 2 * stats::pnorm(abs(z), lower.tail = FALSE) else p
    data.frame(
      term = term,

```

```

      estimate = estimate,
      std.error = se,
      statistic = z,
      p.value = p_out,
      s.value = if (is.finite(p_out) && p_out > 0) -log2(p_out) else NA_real_,
      conf.low = if (is.finite(se)) estimate - 1.96 * se else NA_real_,
      conf.high = if (is.finite(se)) estimate + 1.96 * se else NA_real_,
      method = "delta_method_analytic_probit",
      status = "estimated",
      reason = NA_character_,
      stringsAsFactors = FALSE
    )
  })
do.call(rbind, rows)
}

is_marginaeffects_object <- function(x) {
  inherits(x, c("marginaeffects", "comparisons", "avg_comparisons", "slopes", "avg_slopes",
    ↪ "predictions"))
}

copy_first_ame_column <- function(out, target, candidates) {
  if (!target %in% names(out)) out[[target]] <- NA
  for (candidate in candidates) {
    if (!candidate %in% names(out)) next
    missing_target <- is.na(out[[target]])
    if (any(missing_target)) out[[target]][missing_target] <-
      ↪ out[[candidate]][missing_target]
  }
  out
}

normalize_ame_columns <- function(out) {
  # marginaeffects has used both dotted and snake_case names across versions
  # and model classes. Normalize here before selecting the public/audit schema
  # so uncertainty columns are not silently dropped downstream.
  aliases <- list(
    std.error = c("std_error", "std.err", "Std. Error"),
    statistic = c("z", "z.value", "z_value", "t", "t.value", "t_value"),
    p.value = c("p_value", "p", "Pr(>|z|)", "Pr(>|t|)"),
    s.value = c("s_value"),
    conf.low = c("conf_low", "conf.low", "2.5 %"),
    conf.high = c("conf_high", "conf.high", "97.5 %")
  )
  for (target in names(aliases)) {
    out <- copy_first_ame_column(out, target, aliases[[target]])
  }
  out
}

ame_label_lookup <- function() {
  data.frame(
    term = c(
      "AGE", "SEX", "HH_SIZE",

```

```

"RELIGION", "RELIGION", "RELIGION", "RELIGION", "RELIGION", "RELIGION", "RELIGION",
"SOCIAL_GROUP", "SOCIAL_GROUP", "SOCIAL_GROUP",
"SECTOR",
"DIST_FROM_NEAREST_PRIMARY_CLASS", "DIST_FROM_NEAREST_PRIMARY_CLASS",
  ↪ "DIST_FROM_NEAREST_PRIMARY_CLASS", "DIST_FROM_NEAREST_PRIMARY_CLASS",
"father_educ", "father_educ", "father_educ", "father_educ", "father_educ",
  ↪ "father_educ", "father_educ",
"dmean_num_IS_EDU_FREE", "dmean_num_TUTION_FEE_WAIVED",
  ↪ "dmean_num_RECD_SCHOLARSHIP_STIPEND",
"dmean_num_RECD_TXT_BOOKS", "dmean_num_RECD_STATIONERY",
  ↪ "dmean_num_MID_DAY_MEAL_ETC_RECD", "dmean_num_ENROLLMENT_COST"
),
contrast = c(
  "dY/dX", "Female - Male", "dY/dX",
  "Muslim - Hindu", "Christian - Hindu", "Sikh - Hindu", "Jain - Hindu", "Buddhist -
    ↪ Hindu", "Zoroastrian - Hindu", "Other - Hindu",
  "Scheduled Tribe - Other", "Scheduled Caste - Other", "Other Backward Class - Other",
  "Urban - Rural",
  "1km <= d <2kms - d<1km", "2kms<= d <3kms - d<1km", "3kms <= d <5kms - d<1km",
    ↪ "d>=5kms - d<1km",
  "Literate, no school - Illiterate", "Literate, school < primary - Illiterate",
    ↪ "Primary - Illiterate", "Upper primary - Illiterate", "Secondary - Illiterate",
    ↪ "Higher secondary - Illiterate", "Postsecondary+ - Illiterate",
  "dY/dX", "dY/dX", "dY/dX", "dY/dX", "dY/dX", "dY/dX", "dY/dX"
),
Term = c(
  "Age (years)", "Female (ref: Male)", "Household size",
  "Religion: Muslim (ref: Hindu)", "Religion: Christian", "Religion: Sikh", "Religion:
    ↪ Jain", "Religion: Buddhist", "Religion: Zoroastrian", "Religion: Other",
  "Social group: Scheduled Tribe (ref: Other)", "Social group: Scheduled Caste", "Social
    ↪ group: Other Backward Class",
  "Urban (ref: Rural)",
  "Distance 1-2km (ref: <1km)", "Distance 2-3km", "Distance 3-5km", "Distance > 5km",
  "Father's educ.: Literate, no school (ref: Illiterate)", "Father's educ.: Literate,
    ↪ school < primary", "Father's educ.: Primary", "Father's educ.: Upper primary",
    ↪ "Father's educ.: Secondary", "Father's educ.: Higher secondary", "Father's educ.:
    ↪ Postsecondary+",
  "Educ. free available (ref: No)", "Tuition waiver received", "Scholarship/Stipend
    ↪ received", "Textbook(s) received", "Stationery received", "Mid-day meal, etc.
    ↪ received", "Enrollment cost (Rs.)"
),
stringsAsFactors = FALSE
)
}

ame_label_order <- function() ame_label_lookup()$Term

ame_modelsummary_label <- function(x) {
  x <- as.character(x)
  x <- gsub("\u00d7", ":", x, fixed = TRUE)
  x <- gsub("\\s+:\\s+", ": ", x, perl = TRUE)
  x <- gsub("\\(ref:\\s*", "(ref: ", x, perl = TRUE)
  x
}

```

```

infer_ame_contrast <- function(out) {
  if ("contrast" %in% names(out)) return(as.character(out$contrast))
  contrast <- rep("dY/dX", nrow(out))
  if (!"term" %in% names(out)) return(contrast)
  if ("comparison" %in% names(out)) contrast <- as.character(out$comparison)
  contrast[is.na(contrast) | !nzchar(contrast)] <- "dY/dX"
  contrast
}

ame_factor_base_terms <- function() {
  unique(ame_label_lookup()$term[ame_label_lookup()$contrast != "dY/dX"])
}

normalize_encoded_factor_ame_terms <- function(out) {
  out <- as.data.frame(out, stringsAsFactors = FALSE)
  if (!"term" %in% names(out)) return(out)
  if (!"contrast" %in% names(out)) out$contrast <- infer_ame_contrast(out)

  lookup <- ame_label_lookup()
  factor_terms <- ame_factor_base_terms()
  factor_terms <- factor_terms[order(nchar(factor_terms), decreasing = TRUE)]

  for (i in seq_len(nrow(out))) {
    current_term <- as.character(out$term[[i]])
    current_contrast <- as.character(out$contrast[[i]])
    if (!nzchar(current_term) || (!is.na(current_contrast) && current_contrast != "dY/dX"))
      ↪ next

    direct <- lookup$term == current_term & lookup$contrast == current_contrast
    if (any(direct)) next

    for (base_term in factor_terms) {
      if (!startsWith(current_term, base_term)) next
      level <- trimws(sub(paste0("^", base_term), "", current_term))
      if (!nzchar(level)) next
      candidates <- lookup[lookup$term == base_term, , drop = FALSE]
      contrast_hit <- candidates$contrast[startsWith(candidates$contrast, paste0(level, " -
↪ "))]]
      if (length(contrast_hit)) {
        out$term[[i]] <- base_term
        out$contrast[[i]] <- contrast_hit[[1]]
        break
      }
    }
  }
  out
}

attach_ame_labels <- function(out) {
  out <- as.data.frame(out, stringsAsFactors = FALSE)
  if (!"term" %in% names(out) && "variable" %in% names(out)) out$term <- out$variable
  if (!"contrast" %in% names(out)) out$contrast <- infer_ame_contrast(out)
  out <- normalize_encoded_factor_ame_terms(out)
  lookup <- ame_label_lookup()
  labeled <- merge(out, lookup, by = c("term", "contrast"), all.x = TRUE, sort = FALSE)
}

```



```

labeled$Term[is.na(labeled$Term)] <- labeled$term[is.na(labeled$Term)]
labeled$Term <- factor(labeled$Term, levels = unique(c(ame_label_order(), labeled$Term)))
labeled <- labeled[order(labeled$Term), , drop = FALSE]
labeled$Term <- as.character(labeled$Term)
labeled
}

```

```

modelsummary_margineffects_object <- function(out, native_ame) {
  if (!is_margineffects_object(native_ame)) return(NULL)
  out <- as.data.frame(out, stringsAsFactors = FALSE)
  required <- c("Term", "estimate", "std.error", "statistic", "p.value", "conf.low",
    ↪ "conf.high")
  if (!all(required %in% names(out))) return(native_ame)

```

```

  ms <- out[, required, drop = FALSE]
  ms$term <- ame_modelsummary_label(ms$Term)
  ms$contrast <- ""
  ms$Term <- NULL
  # Preserve the margineffects class and non-structural attributes so
  # modelsummary can use its native margineffects/tidy pathway. We still
  # hand modelsummary the labeled, ordered rows that are validated for the
  # public table, because passing the raw object caused modelsummary to reorder
  # contrasts and display labels against the wrong estimates.
  native_attributes <- attributes(native_ame)
  structural_attributes <- c("names", "row.names", "class")
  for (nm in setdiff(names(native_attributes), structural_attributes)) {
    attr(ms, nm) <- native_attributes[[nm]]
  }
  class(ms) <- unique(c(class(native_ame), class(ms)))
  ms
}

```

```

#' format ame results
#'

```

```

format_ame_results <- function(ame_results) {
  native_ame <- ame_results
  out <- tibble::as_tibble(ame_results)
  if (!"term" %in% names(out) && "variable" %in% names(out)) out$term <- out$variable
  if (!"contrast" %in% names(out)) out$contrast <- infer_ame_contrast(as.data.frame(out))
  out <- normalize_ame_columns(out)
  if (!"method" %in% names(out)) out$method <- "autodiff"
  if (!"status" %in% names(out)) out$status <- "estimated"
  if (!"reason" %in% names(out)) out$reason <- NA_character_
  labeled_terms <- unique(ame_label_lookup()$term)
  use_labeled_schema <- any(out$term %in% labeled_terms) ||
    any(grepl("^dmean_num_", out$term ||% character(), perl = TRUE)) ||
    any((out$contrast ||% "dY/dX") != "dY/dX", na.rm = TRUE)

  if (isTRUE(use_labeled_schema)) {
    out <- attach_ame_labels(out)
    required <- c(
      "Term", "term", "contrast", "estimate", "std.error", "statistic", "p.value",
      ↪ "s.value",
      "conf.low", "conf.high", "method", "status", "reason"
    )
  }
}

```

```

    )
  } else {
    required <- c(
      "term", "estimate", "std.error", "statistic", "p.value",
      "s.value", "conf.low", "conf.high", "method", "status", "reason"
    )
  }

  for (nm in setdiff(required, names(out))) out[[nm]] <- NA
  if ("p.value" %in% names(out) && "s.value" %in% names(out)) {
    missing_s <- is.na(out$s.value) & is.finite(out$p.value) & out$p.value > 0
    out$s.value[missing_s] <- -log2(out$p.value[missing_s])
  }
  out <- out[, required, drop = FALSE]
  if (is_margineffects_object(native_ame)) {
    attr(out, "marginaleffects_object") <- modelsummary_marginaleffects_object(out,
      ↪ native_ame)
  }
  out
}

#' save ame results
#'
save_ame_results <- function(ame_results, path =
  ↪ "outputs/tables/diagnostics/ame_results.csv") {
  readr::write_csv(tibble::as_tibble(ame_results), path); path
}

```

Cascading fuzzy district record linkage

Source: R/districts/fuzzy_join_districts.R

```

# The functions below implement the current cascading fuzzy-match architecture.
# sample-end marker appears near the end of this file for coding-sample extraction.

#' fuzzy join sequence
#'
fuzzy_join_sequence <- function(df1, df2, dist1, state1, dist2, state2, methods, thresholds,
  ↪ mode = "full") {
  df1_id <- df1 |> dplyr::mutate(.id1 = dplyr::row_number())
  df2_id <- df2 |> dplyr::mutate(.id2 = dplyr::row_number())
  df1_curr <- df1_id; df2_curr <- df2_id; matched_all <- tibble::tibble()
  for (i in seq_along(methods)) {
    full_j <- fuzzyjoin::stringdist_join(df1_curr, df2_curr, by = stats::setNames(c(dist2,
  ↪ state2), c(dist1, state1)), mode = mode, method = methods[i], max_dist = thresholds[i],
    ↪ distance_col = "dist")
    matched_i <- full_j |> dplyr::filter(!is.na(.id1), !is.na(.id2)) |>
    ↪ dplyr::group_by(.id1) |> dplyr::slice_min(dist, n = 1, with_ties = FALSE) |>
    ↪ dplyr::ungroup() |> dplyr::group_by(.id2) |> dplyr::slice_min(dist, n = 1, with_ties =
    ↪ FALSE) |> dplyr::ungroup()
    matched_all <- dplyr::bind_rows(matched_all, matched_i)
    df1_curr <- df1_curr |> dplyr::filter(!(.id1 %in% matched_i$.id1)); df2_curr <- df2_curr
    ↪ |> dplyr::filter(!(.id2 %in% matched_i$.id2))
  }
  list(joined = matched_all, unmatched_df1 = df1_curr |> dplyr::select(-.id1), unmatched_df2
    ↪ = df2_curr |> dplyr::select(-.id2))
}

```

```

}

#' unmatched rows
#'
#' @return A data frame of unmatched rows when the join map carries that attribute.
unmatched_rows <- function(join_map, ...) {
  attr(join_map, "unmatched_rows") %||% join_map[0, , drop = FALSE]
}

#' merge dfs into tracker
#'
#' Cascading district matcher. `df_names` is a named list (or a
#' data frame) whose elements carry source-specific district/state names. For
#' each source we try every requested tracker year in order, keep one-to-one
#' canonical name matches, and then fuzzy-match remaining rows within state.
#'
#' @return A list with `joined_df`, `unmatched_df`, and `flagged_df`, matching
#' the current matcher object contract.
merge_dfs_into_tracker <- function(df_names, tracker = NULL, years_of_interest = c("2001",
  ↪ "2007", "2008", "2017", "2018", "2020"), flag = TRUE, ...) {
  if (is.null(tracker)) return(list(joined_df = safe_df(df_names), unmatched_df =
  ↪ data.frame(), flagged_df = data.frame()))
  tracker <- safe_df(tracker)
  if (!nrow(tracker)) return(list(joined_df = data.frame(), unmatched_df =
  ↪ safe_df(df_names), flagged_df = data.frame()))
  if (!".tracker_row" %in% names(tracker)) tracker$.tracker_row <- seq_len(nrow(tracker))
  xs <- if (inherits(df_names, "data.frame")) list(source = df_names) else
  ↪ as_input_list(df_names)

  joined_all <- list()
  unmatched_all <- list()
  flagged_all <- list()
  for (source_name in names(xs)) {
    source <- safe_df(xs[[source_name]])
    if (!nrow(source)) next
    source$.source_row <- seq_len(nrow(source))
    source$.source_name <- source_name
    source <- add_source_name_keys(source)
    if (!all(c(".source_state_key", ".source_district_key") %in% names(source))) {
      unmatched_all[[source_name]] <- source
      next
    }

    source$.matched <- FALSE
    for (yr in years_of_interest) {
      sfx <- substr(as.character(yr), 3, 4)
      state_col <- paste0("state_", sfx)
      district_col <- paste0("district_", sfx)
      if (!all(c(state_col, district_col) %in% names(tracker))) next
      candidate <- tracker[c(".tracker_row", state_col, district_col)]
      candidate$.tracker_state_key <- canon(candidate[[state_col]])
      candidate$.tracker_district_key <- canon(candidate[[district_col]])
      exact <- merge(
        source[!source$.matched, , drop = FALSE],
        candidate,

```

```

    by.x = c(".source_state_key", ".source_district_key"),
    by.y = c(".tracker_state_key", ".tracker_district_key"),
    all.x = FALSE,
    all.y = FALSE
  )
  if (!nrow(exact)) next
  exact <- exact[!duplicated(exact$.source_row) & !duplicated(exact$.tracker_row), ,
  drop = FALSE]
  if (!nrow(exact)) next
  exact$match_year <- as.character(yr)
  exact$match_status <- "exact_name"
  joined_all[[paste(source_name, yr, sep = "_")] <- exact
  source$.matched[source$.source_row %in% exact$.source_row] <- TRUE
}

if (any(!source$.matched)) {
  fuzzy <- fuzzy_match_remaining_to_tracker(source[!source$.matched, , drop = FALSE],
  tracker, years_of_interest)
  if (nrow(fuzzy)) {
    joined_all[[paste0(source_name, "_fuzzy")] <- fuzzy
    source$.matched[source$.source_row %in% fuzzy$.source_row] <- TRUE
    if (flag) flagged_all[[source_name]] <- fuzzy[fuzzy$possible_false_positive %in%
    TRUE, , drop = FALSE]
  }
}
unmatched_all[[source_name]] <- source[!source$.matched, , drop = FALSE]
}
list(
  joined_df = safe_bind_rows(joined_all),
  unmatched_df = safe_bind_rows(unmatched_all),
  flagged_df = safe_bind_rows(flagged_all)
)
}

add_source_name_keys <- function(source) {
  state <- first_col(source, c("state", "state_01", "state_07", "state_08", "state_17",
  "state_18", "state_20", "state_0708", "state_1718"))
  district <- first_col(source, c("district", "district_01", "district_07", "district_08",
  "district_17", "district_18", "district_20", "district_0708", "district_1718",
  "district_name"))
  if (!is.null(state)) source$.source_state_key <- canon(source[[state]])
  if (!is.null(district)) source$.source_district_key <- canon(source[[district]])
  source
}

fuzzy_match_remaining_to_tracker <- function(source, tracker, years_of_interest) {
  out <- list()
  for (yr in years_of_interest) {
    sfx <- substr(as.character(yr), 3, 4)
    state_col <- paste0("state_", sfx)
    district_col <- paste0("district_", sfx)
    if (!all(c(state_col, district_col) %in% names(tracker))) next
    candidate <- tracker[c(".tracker_row", state_col, district_col)]
    candidate$.tracker_state_key <- canon(candidate[[state_col]])
    candidate$.tracker_district_key <- canon(candidate[[district_col]])
  }
}

```

```

    for (i in seq_len(nrow(source))) {
      pool <- candidate[candidate$.tracker_state_key == source$.source_state_key[[i]], ,
        drop = FALSE]
      if (!nrow(pool)) next
      dist <- utils::adist(source$.source_district_key[[i]], pool$.tracker_district_key,
        ignore.case = TRUE)[1, ]
      best <- which.min(dist)
      if (!length(best) || !is.finite(dist[[best]]) || dist[[best]] > 2) next
      row <- cbind(source[i, , drop = FALSE], pool[best, , drop = FALSE])
      row$match_year <- as.character(yr)
      row$match_status <- "fuzzy_name"
      row$dist <- dist[[best]]
      row$possible_false_positive <- dist[[best]] > 0
      out[[length(out) + 1L]] <- row
    }
  }
  x <- safe_bind_rows(out)
  if (nrow(x)) x <- x[!duplicated(x$.source_row) & !duplicated(x$.tracker_row), , drop =
    FALSE]
  x
}

#' join year to tracker
#'
join_year_to_tracker <- function(source, tracker, years_of_interest, ...) {
  merge_dfs_into_tracker(source, tracker = tracker, years_of_interest = years_of_interest,
    ...)
}

#' join all district sources
#'
join_all_district_sources <- function(df_names, tracker, years_of_interest, ...) {
  merge_dfs_into_tracker(df_names, tracker = tracker, years_of_interest = years_of_interest,
    ...)
}

#' flag many to many matches
#'
flag_many_to_many_matches <- function(join_map, source_col = NULL, tracker_col = NULL) {
  join_map <- safe_df(join_map)
  if (!nrow(join_map)) {
    join_map$many_to_many <- logical()
    join_map$many_to_many_type <- character()
    return(join_map)
  }
  source_col <- source_col %||% first_col(join_map, c(".source_row", "source_row",
    "source_key", "source_id"))
  tracker_col <- tracker_col %||% first_col(join_map, c(".tracker_row", "tracker_row",
    "tracker_key", "district_panel_id"))
  if (is.null(source_col) || is.null(tracker_col)) {
    join_map$many_to_many <- FALSE
    join_map$many_to_many_type <- "not_evaluable_missing_keys"
    return(join_map)
  }
  source_key <- canon(join_map[[source_col]])

```

```

tracker_key <- canon(join_map[[tracker_col]])
source_n <- ave(rep(1L, nrow(join_map)), source_key, FUN = length)
tracker_n <- ave(rep(1L, nrow(join_map)), tracker_key, FUN = length)
source_dup <- source_n > 1L & !is.na(source_key) & nzchar(source_key)
tracker_dup <- tracker_n > 1L & !is.na(tracker_key) & nzchar(tracker_key)
join_map$n_source_matches <- as.integer(source_n)
join_map$n_tracker_matches <- as.integer(tracker_n)
join_map$many_to_many <- source_dup & tracker_dup
join_map$many_to_many_type <- ifelse(
  source_dup & tracker_dup, "many_to_many",
  ifelse(source_dup, "one_source_to_many_tracker",
    ifelse(tracker_dup, "many_source_to_one_tracker", "one_to_one"))
)
join_map
}

#' score candidate matches
#'
score_candidate_matches <- function(candidates) {
  if (!"dist" %in% names(candidates)) candidates$dist <- 0
  candidates$score <- -num(candidates$dist)
  candidates
}

#' fuzzy join districts
#'
fuzzy_join_districts <- function(district_tracker, district_keys_2001, district_keys_2007,
  ↪ district_keys_2017, district_keys_2020, cfg) {
  tracker <- safe_df(district_tracker)
  if (nrow(tracker) && all(c("state_01", "district_01", "state_07", "district_07",
    ↪ "state_17", "district_17", "state_20", "district_20") %in% names(tracker))) {
    tracker$.tracker_row <- seq_len(nrow(tracker))
    tracker$source <- "harmonization_crosswalk"
    tracker$match_status <- "harmonization_crosswalk_row"
    tracker$possible_false_positive <- FALSE
    tracker$many_to_many <- FALSE
    attr(tracker, "unmatched_rows") <- tracker[0, , drop = FALSE]
    attr(tracker, "possible_false_positives") <- tracker[0, , drop = FALSE]
    attr(tracker, "many_to_many_cases") <- tracker[0, , drop = FALSE]
    return(tracker)
  }

  out <- safe_bind_rows(Map(function(keys, source) {
    if (!nrow(keys)) return(data.frame())
    keys$source <- source
    keys$match_status <- "source_key_unmatched"
    keys$possible_false_positive <- FALSE
    keys$many_to_many <- FALSE
    keys
  }, list(district_keys_2001, district_keys_2007, district_keys_2017, district_keys_2020),
  ↪ c("2001", "2007", "2017", "2020")))
  attr(out, "unmatched_rows") <- out
  attr(out, "possible_false_positives") <- out[out$possible_false_positive, , drop = FALSE]
  attr(out, "many_to_many_cases") <- out[out$many_to_many, , drop = FALSE]
  out

```

```
}
```

Contiguity-based spatial weights

Source: R/diagnostics/diagnose_spatial_weights.R

```
#' build spatial weights
#'
```

#' Build the rook-contiguity spatial weights used by the legacy Rmd.

#'

#' The legacy chunk first removed rows with missing analysis values, then called

#' `poly2nb(queen = FALSE)`, `nb2mat(style = "B")`, and `nb2listw(style = "W")`.

#' The target stores all three objects so downstream diagnostics can reproduce

#' both the binary adjacency matrix and the row-standardized listw object.

#'

#' @return A list containing neighbor, matrix, and listw objects, or an explicit

#' inactive status when geometry is unavailable.

```
build_spatial_weights <- function(district_panel, cfg) {
  if (!inherits(district_panel, "sf")) {
    return(list(status = "out_of_active_pipeline", reason = "Requires sf geometry."))
  }
  need_pkg("spdep", "spatial weights")
  need_pkg("sf", "spatial weights")

  row_index <- spatial_weight_final_panel_rows(district_panel)
  coverage <- length(row_index) / max(nrow(district_panel), 1L)
  if (!is.finite(coverage) || coverage < 0.75) {
    return(list(
      status = "out_of_active_pipeline",
      reason = paste0(
        "Requires validated non-empty geometry for at least 75% of district-panel rows;
        ↪ current coverage is ",
        round(100 * coverage, 1), "%."
      )
    ))
  }

  weights <- build_spatial_weights_for_rows(district_panel, row_index, queen = FALSE)
  weights$coverage <- coverage
  weights$panel_scope <- "current_final_matched_panel_non_empty_geometry"
  weights
}
```

#' rows used by final-panel spatial diagnostics

#'

#' Spatial diagnostics operate on the active `district\_panel` target, not on the

#' legacy exploratory geometry object. The final-panel scope is therefore all

#' rows in the current matched panel with non-empty geometry.

```
spatial_weight_final_panel_rows <- function(district_panel) {
  if (!inherits(district_panel, "sf")) return(integer())
  geom <- sf::st_geometry(district_panel)
  which(!sf::st_is_empty(geom))
}
```

#' build spatial weights for selected district-panel rows

#'

#' @return A list with nb, binary matrix, and row-standardized listw objects.

```

build_spatial_weights_for_rows <- function(district_panel, rows, queen = FALSE) {
  if (!inherits(district_panel, "sf")) {
    return(list(status = "out_of_active_pipeline", reason = "Requires sf geometry."))
  }
  need_pkg("spdep", "spatial weights")
  need_pkg("sf", "spatial weights")

  rows <- as.integer(rows)
  rows <- rows[is.finite(rows) & rows >= 1L & rows <= nrow(district_panel)]
  if (!length(rows)) {
    return(list(status = "out_of_active_pipeline", reason = "No rows with usable
      ↪ geometry."))
  }
  geom <- sf::st_geometry(district_panel)
  rows <- rows[!sf::st_is_empty(geom[rows])]
  if (!length(rows)) {
    return(list(status = "out_of_active_pipeline", reason = "Selected rows have empty
      ↪ geometry."))
  }

  spatial_warnings <- character()
  capture_expected_spatial_warning <- function(expr) {
    withCallingHandlers(
      expr,
      warning = function(w) {
        msg <- conditionMessage(w)
        expected <- grepl("no neighbours|sub-graphs", msg, ignore.case = TRUE)
        if (expected) {
          spatial_warnings <-<- unique(c(spatial_warnings, msg))
          invokeRestart("muffleWarning")
        }
      }
    )
  }

  panel <- district_panel[rows, , drop = FALSE]
  nb <- capture_expected_spatial_warning(spdep::poly2nb(panel, queen = queen))
  W <- capture_expected_spatial_warning(spdep::nb2mat(nb, style = "B", zero.policy = TRUE))
  listw <- capture_expected_spatial_warning(spdep::nb2listw(nb, style = "W", zero.policy =
    ↪ TRUE))

  out <- list(
    status = "constructed",
    contiguity = if (isTRUE(queen)) "queen" else "rook",
    style = "W",
    matrix_style = "B",
    zero_policy = TRUE,
    row_index = rows,
    nb = nb,
    W = W,
    listw = listw,
    neighbor_counts = lengths(nb),
    n = length(rows),
    n_islands = sum(lengths(nb) == 0L),
    mean_neighbors = mean(lengths(nb)),
  )

```



```

    warnings = spatial_warnings
  )
  class(out) <- c("emi_spatial_weights", class(out))
  out
}

#' diagnose spatial weights
#'
diagnose_spatial_weights <- function(district_panel, spatial_weights, cfg) {
  if (!inherits(district_panel, "sf")) {
    return(data.frame(
      diagnostic = "spatial_weights",
      status = "out_of_active_pipeline",
      reason = "Requires sf geometry.",
      stringsAsFactors = FALSE
    ))
  }

  comparison <- compare_rook_queen_contiguity(district_panel)
  base <- data.frame(
    diagnostic = "spatial_weights",
    status = spatial_weights$status %||% "constructed",
    contiguity = spatial_weights$contiguity %||% NA_character_,
    style = spatial_weights$style %||% NA_character_,
    matrix_style = spatial_weights$matrix_style %||% NA_character_,
    n = spatial_weights$n %||% NA_real_,
    n_islands = spatial_weights$n_islands %||% NA_real_,
    mean_neighbors = spatial_weights$mean_neighbors %||% NA_real_,
    panel_scope = spatial_weights$panel_scope %||%
      ↪ "current_final_matched_panel_non_empty_geometry",
    warnings = paste(spatial_weights$warnings %||% attr(spatial_weights, "spatial_warnings")
      ↪ %||% character(), collapse = "; "),
    stringsAsFactors = FALSE
  )
  attr(base, "rook_queen_comparison") <- add_spatial_weight_reference(comparison)
  attr(base, "legacy_reference") <- spatial_weight_reference()
  base
}

#' compare rook and queen contiguity
#'
#' The legacy Rmd commented out an sfExtras rook-vs-queen check and recorded
#' nearly identical mean neighbor counts (rook 4.780165, queen 4.783471) plus
#' similar run times. To avoid an extra sfExtras dependency, this project uses
#' the same `spdep::poly2nb()` implementation used by the final rook weights and
#' records elapsed time and average neighbor counts for both choices.
compare_rook_queen_contiguity <- function(district_panel) {
  if (!inherits(district_panel, "sf")) return(tibble::tibble())
  need_pkg("spdep", "rook/queen contiguity comparison")
  need_pkg("sf", "rook/queen contiguity comparison")
  panel <- district_panel[spatial_weight_final_panel_rows(district_panel), , drop = FALSE]
  if (!nrow(panel)) return(tibble::tibble())

  one <- function(queen) {
    warnings <- character()

```

```

elapsed <- system.time({
  nb <- withCallingHandlers(
    spdep::poly2nb(panel, queen = queen),
    warning = function(w) {
      msg <- conditionMessage(w)
      if (grepl("no neighbours|sub-graphs", msg, ignore.case = TRUE)) {
        warnings <-<- unique(c(warnings, msg))
        invokeRestart("muffleWarning")
      }
    }
  )
})[["elapsed"]]
tibble::tibble(
  contiguity = if (isTRUE(queen)) "queen" else "rook",
  n = length(nb),
  mean_neighbors = mean(lengths(nb)),
  n_islands = sum(lengths(nb) == 0L),
  panel_scope = "current_final_matched_panel_non_empty_geometry",
  elapsed_seconds = unname(elapsed),
  warnings = paste(warnings, collapse = "; ")
)
}

safe_bind_rows(list(one(FALSE), one(TRUE)))
}

spatial_weight_reference <- function() {
  data.frame(
    contiguity = c("rook", "queen"),
    legacy_method = c(
      "sfExtras::st_rook() timing comment; final weights use spdep::poly2nb(queen = FALSE)",
      "sfExtras::st_queen() timing comment; benchmark uses spdep::poly2nb(queen = TRUE)"
    ),
    legacy_mean_neighbors = c(4.780165, 4.783471),
    legacy_elapsed_note = c("legacy comment recorded similar run time to queen", "legacy
      ↪ comment recorded similar run time to rook"),
    interpretation = "Current means are intentionally computed on the active final matched
      ↪ panel with non-empty geometry; they may differ from legacy exploratory sfExtras
      ↪ objects, but they now match the panel used for current spatial diagnostics.",
    stringsAsFactors = FALSE
  )
}

add_spatial_weight_reference <- function(comparison) {
  comparison <- as.data.frame(comparison, stringsAsFactors = FALSE)
  if (!nrow(comparison)) return(comparison)
  ref <- spatial_weight_reference()[c("contiguity", "legacy_mean_neighbors")]
  out <- merge(comparison, ref, by = "contiguity", all.x = TRUE, sort = FALSE)
  out$mean_neighbor_delta_from_legacy <- out$mean_neighbors - out$legacy_mean_neighbors
  out$pct_delta_from_legacy <- 100 * out$mean_neighbor_delta_from_legacy /
    ↪ out$legacy_mean_neighbors
  out
}

```

```

#' summarize islands
#'
summarize_islands <- function(spatial_weights) {
  if (!inherits(spatial_weights, "emi_spatial_weights")) return(tibble::tibble())
  islands <- which(spatial_weights$neighbor_counts == 0L)
  tibble::tibble(row_index = spatial_weights$row_index[islands], n_neighbors = 0L)
}

#' summarize neighbor counts
#'
summarize_neighbor_counts <- function(spatial_weights) {
  if (!inherits(spatial_weights, "emi_spatial_weights")) return(tibble::tibble())
  tibble::tibble(
    row_index = spatial_weights$row_index,
    n_neighbors = spatial_weights$neighbor_counts
  )
}

#' save spatial weight diagnostics
#'
save_spatial_weight_diagnostics <- function(diagnostics, dir =
  ↪ "outputs/diagnostics/extended/spatial") {
  dir.create(dir, recursive = TRUE, showWarnings = FALSE)
  paths <- c(
    summary = write_diagnostic_csv(as.data.frame(diagnostics), file.path(dir,
      ↪ "spatial_weights_summary.csv")),
    rook_queen_comparison = write_diagnostic_csv(attr(diagnostics, "rook_queen_comparison")
      ↪ %||% data.frame(), file.path(dir, "rook_queen_contiguity_comparison.csv")),
    legacy_reference = write_diagnostic_csv(attr(diagnostics, "legacy_reference") %||%
      ↪ data.frame(), file.path(dir, "spatial_weights_legacy_reference.csv"))
  )
  output_manifest(paths)
}

```

Reusable IV specification and estimation

Source: R/iv/build_iv_formulas.R

```

#' make iv formula
#'
make_iv_formula <- function(dep, endog, instruments, controls = NULL, fixed_effects = NULL)
  ↪ {
  stats::as.formula(paste(
    dep,
    "~",
    paste(c(endog, controls, fixed_effects), collapse = " + "),
    "|",
    paste(c(instruments, controls, fixed_effects), collapse = " + ")
  ))
}

baseline_iv_controls <- function() {
  c(
    "consumption_0708", "gini_cons_0708",
    "pct_urban", "avg_hh_size", "dependency_ratio",
    "pct_fem_head",

```

```

    "pct_hindu", "pct_muslim",
    "pct_st", "pct_sc", "pct_obc",
    "pct_small_land", "pct_medium_land", "pct_large_land",
    "pct_head_lit_to_primary", "pct_head_secondary_plus"
  )
}

#' build iv formulas
#'
build_iv_formulas <- function(cfg) {
  controls <- baseline_iv_controls()
  list(
    consumption = make_iv_formula(
      "consumption_pct_change",
      "EMIE",
      "wavg_ling_degrees",
      controls
    ),
    gini = make_iv_formula(
      "gini_change",
      "EMIE",
      "wavg_ling_degrees",
      controls
    )
  )
}

#' build baseline 2sls formula
#'
build_baseline_2sls_formula <- function(...) make_iv_formula(...)

```

Multicollinearity and spatial autocorrelation diagnostics

Source: R/diagnostics/diagnose_multicollinearity.R

```

#' diagnose multicollinearity
#'
#' @return A data frame with design-matrix condition diagnostics.
diagnose_multicollinearity <- function(district_panel, iv_models, cfg) {
  model <- if (is.list(iv_models) && length(iv_models)) iv_models[[1]] else iv_models
  out <- tryCatch({
    X <- stats::model.matrix(model)
    data.frame(
      test = "kappa",
      n = nrow(X),
      rank = qr(X)$rank,
      columns = ncol(X),
      kappa = kappa(X, exact = TRUE),
      status = "estimated",
      reason = NA_character_,
      stringsAsFactors = FALSE
    )
  }, error = function(e) {
    data.frame(
      test = "kappa",
      n = NA_integer_,

```

```

    rank = NA_integer_,
    columns = NA_integer_,
    kappa = NA_real_,
    status = "out_of_active_pipeline",
    reason = conditionMessage(e),
    stringsAsFactors = FALSE
  )
})
out
}

#' compute design matrix rank
#'
compute_design_matrix_rank <- function(formula, data) {
  X <- stats::model.matrix(formula, data = data); tibble::tibble(rank = qr(X)$rank, ncol =
  ↪ ncol(X))
}

#' compute kappa
#'
compute_kappa <- function(formula, data) {
  X <- stats::model.matrix(formula, data = data); kappa(X, exact = TRUE)
}

#' compute vif if applicable
#'
compute_vif_if_applicable <- function(model) {
  if (requireNamespace("car", quietly = TRUE)) car::vif(model) else NULL
}

#' save multicollinearity diagnostics
#'
save_multicollinearity_diagnostics <- function(diagnostics) {
  diagnostics
}

```

Automated figure generation

Source: R/output/make_figures.R

```

figure_spec <- function(name, file, title, subtitle = NULL, kind = "status", variable =
  ↪ NULL, sources = NULL, inputs = NULL) {
  out <- list(
    name = name,
    file = file,
    title = title,
    subtitle = subtitle,
    kind = kind,
    variable = variable,
    sources = sources,
    inputs = inputs
  )
}

has_sf_geometry <- function(x) {
  inherits(x, "sf") && !is.null(attr(x, "sf_column")) && attr(x, "sf_column") %in% names(x)
}

```

```

}

sf_geometry_coverage <- function(x) {
  if (!has_sf_geometry(x) || !nrow(x)) return(0)
  mean(!sf::st_is_empty(sf::st_geometry(x)))
}

require_final_figure_inputs <- function(district_panel, cfg, required_variables) {
  if (!identical(cfg$mode, "final")) return(invisible(TRUE))
  missing_vars <- setdiff(required_variables, names(as.data.frame(district_panel)))
  if (length(missing_vars)) {
    stop(
      "Final figure generation requires mapped variables. Missing variables: ",
      paste(missing_vars, collapse = ", "),
      call. = FALSE
    )
  }
  if (!has_sf_geometry(district_panel)) {
    stop("Final map generation requires an sf district_panel with validated geometry.",
      call. = FALSE)
  }
  coverage <- sf_geometry_coverage(district_panel)
  if (!is.finite(coverage) || coverage < 0.75) {
    stop(
      "Final map generation requires a validated geometry join covering at least 75% of
      ↪ district-panel rows; current coverage is ",
      round(100 * coverage, 1),
      "%.",
      call. = FALSE
    )
  }
  invisible(TRUE)
}

#' make figures
#'
#' @return A named list of figure specifications consumed by save_figures().
make_figures <- function(district_panel, raw_ilo_figures, cfg, boundaries_2020 = NULL) {
  required_variables <- c(
    "emie_2007",
    "consumption_growth_pct",
    "pucca_share_2007",
    "head_secondary_plus_2007",
    "region",
    "wavg_ling_degrees"
  )

  out <- list(
    fig_ilo_trends = figure_spec(
      "fig_ilo_trends",
      "fig_ilo_trends.png",
      "ILO labor market indicators",
      "Composed from the archived ILO figure assets.",
      kind = "ilo_collage",
      sources = raw_ilo_figures
    )
  )
}

```

```

    ),
    district_carveouts_shifts = figure_spec(
      "district_carveouts_shifts",
      "district_carveouts_shifts.png",
      "District carve-outs and shifts",
      kind = "district_carveouts_shifts"
    )
  )

missing_vars <- setdiff(required_variables, names(as.data.frame(district_panel)))
geometry_ok <- has_sf_geometry(district_panel) &&
↪ is.finite(sf_geometry_coverage(district_panel)) && sf_geometry_coverage(district_panel)
↪ >= 0.75
maps_available <- !length(missing_vars) && geometry_ok

map_specs <- list(
  map_emi_exposure = figure_spec("map_emi_exposure", "map_emi_exposure.png", "EMI
    ↪ Exposure", kind = if (maps_available) "map" else "status", variable = "emie_2007"),
  map_consumption_growth = figure_spec("map_consumption_growth",
    ↪ "map_consumption_growth.png", "% Change in Consumption", kind = if (maps_available)
    ↪ "map" else "status", variable = "consumption_growth_pct"),
  map_pucca = figure_spec("map_pucca", "map_pucca.png", "% Pucca Homes", kind = if
    ↪ (maps_available) "map" else "status", variable = "pucca_share_2007"),
  map_education = figure_spec("map_education", "map_education.png", "% HH Head w/ Sec.+",
    ↪ kind = if (maps_available) "map" else "status", variable =
    ↪ "head_secondary_plus_2007"),
  map_region = figure_spec("map_region", "map_region.png", "Region", kind = if
    ↪ (maps_available) "map" else "status", variable = "region"),
  map_linguistic_distance = figure_spec("map_linguistic_distance",
    ↪ "map_linguistic_distance.png", "Linguistic Distance", kind = if (maps_available)
    ↪ "map" else "status", variable = "wavg_ling_degrees"),
  collage_main_maps = figure_spec(
    "collage_main_maps",
    "collage_main_maps.png",
    "Main district-level map inputs",
    kind = "collage",
    inputs = c("map_emi_exposure", "map_consumption_growth", "map_pucca", "map_education")
  ),
  collage_iv_region_maps = figure_spec(
    "collage_iv_region_maps",
    "collage_iv_region_maps.png",
    "Instrument and region map inputs",
    kind = "collage",
    inputs = c("map_region", "map_linguistic_distance")
  )
)

if (!maps_available && !identical(cfg$mode, "final")) {
  # Draft-mode diagnostics live outside outputs/figures/main and are explicitly
  # labeled as diagnostics by figure_output_dir().
  map_specs <- map_specs[c("map_emi_exposure", "map_consumption_growth")]
}

map_input_failures <- character()
if (!maps_available && identical(cfg$mode, "final")) {

```

```

map_input_failures <- c(
  if (length(missing_vars)) paste0("Missing map variables: ", paste(missing_vars,
    ↪ collapse = ", ")),
  paste0("Geometry coverage: ", round(100 * sf_geometry_coverage(district_panel), 1),
    ↪ "%")
)
}

out <- c(out, map_specs)

if (length(map_input_failures)) {
  attr(out, "map_input_failures") <- map_input_failures
}
attr(out, "district_panel") <- district_panel
attr(out, "boundaries_2020") <- boundaries_2020
out
}

#' make ilo trends figure
#'
#' @return A vector of source image paths.
make_ilo_trends_figure <- function(raw_ilo_figures) {
  raw_ilo_figures
}

```

Publication-quality regression and summary tables

Source: R/output/make_tables.R

```

table_status_failures <- function(x, cfg, label) {
  df <- as.data.frame(x)
  ok_status <- c("mapped", "estimated")
  if (!is_final_mode(cfg) || !"status" %in% names(df)) return(character())
  bad <- !is.na(df$status) & !df$status %in% ok_status
  if (!any(bad)) return(character())
  reasons <- character()
  if ("reason" %in% names(df)) reasons <-
    ↪ unique(stats::na.omit(as.character(df$reason[bad])))
  suffix <- if (length(reasons)) paste0(" Reasons: ", paste(reasons, collapse = "; ")) else
    ↪ ""
  paste0("Final table generation requires completed model output for ", label, ".", suffix)
}

public_numeric_stats <- function(df, meta, cost_vars = character(), count_vars =
  ↪ character()) {
  df <- as.data.frame(df)
  rows <- lapply(seq_len(nrow(meta)), function(i) {
    v <- meta$var[[i]]
    if (!v %in% names(df)) return(NULL)
    x <- suppressWarnings(as.numeric(as.character(df[[v]])))
    ok <- is.finite(x)
    if (!any(ok)) return(NULL)
    stats <- c(
      Min = min(x[ok], na.rm = TRUE),
      `1Q` = unname(stats::quantile(x[ok], .25, na.rm = TRUE)),
      Med = stats::median(x[ok], na.rm = TRUE),

```



```

    `3Q` = unname(stats::quantile(x[ok], .75, na.rm = TRUE)),
    Max = max(x[ok], na.rm = TRUE),
    Mean = mean(x[ok], na.rm = TRUE),
    SD = stats::sd(x[ok], na.rm = TRUE)
  )
  z <- data.frame(var = v, label = meta$label[[i]], N = sum(ok), t(stats), check.names =
↪ FALSE, stringsAsFactors = FALSE)
  if ("desc" %in% names(meta)) z$desc <- meta$desc[[i]]
  z
})
out <- safe_bind_rows(rows)
if (!nrow(out)) return(data.frame(var = character(), label = character(), N = integer()))
for (nm in intersect(c("Min", "1Q", "Med", "3Q", "Max", "Mean", "SD"), names(out))) {
  x <- suppressWarnings(as.numeric(out[[nm]]))
  out[[nm]] <- ifelse(
    out$var %in% count_vars,
    formatC(round(x), format = "f", big.mark = ",", digits = 0),
    ifelse(out$var %in% cost_vars, formatC(x, format = "f", big.mark = ",", digits = 2),
↪ sprintf("%.2f", x))
  )
}
out
}

public_categorical_stats <- function(df, meta) {
  df <- as.data.frame(df)
  rows <- lapply(seq_len(nrow(meta)), function(i) {
    v <- meta$var[[i]]
    if (!v %in% names(df)) return(NULL)
    x <- df[[v]]
    x <- x[!is.na(x)]
    if (!length(x)) return(NULL)
    freq <- table(x)
    data.frame(
      var = v,
      label = meta$label[[i]],
      N = length(x),
      Values = paste(names(freq), collapse = ", "),
      Mode = names(freq)[which.max(freq)],
      `% Mode` = round(max(freq) / sum(freq) * 100, 1),
      `Least Freq.` = names(freq)[which.min(freq)],
      `% Least Freq.` = round(min(freq) / sum(freq) * 100, 1),
      check.names = FALSE,
      stringsAsFactors = FALSE
    )
  })
  out <- safe_bind_rows(rows)
  if (!nrow(out)) data.frame(var = character(), label = character(), N = integer()) else out
}

summary_group_row <- function(label, columns) {
  row <- as.list(stats::setNames(rep(NA_character_, length(columns)), columns))
  if ("var" %in% columns) row$var <- paste0(".group_", gsub("[^A-Za-z0-9]+", "_", label))
  if ("label" %in% columns) row$label <- label
  as.data.frame(row, check.names = FALSE, stringsAsFactors = FALSE)
}

```

```

}

insert_summary_group <- function(df, label, before_var) {
  if (!nrow(df) || !"var" %in% names(df)) return(df)
  pos <- match(before_var, df$var)
  if (is.na(pos)) return(df)
  rbind(
    if (pos > 1L) df[seq_len(pos - 1L), , drop = FALSE] else df[0L, , drop = FALSE],
    summary_group_row(label, names(df)),
    df[pos:nrow(df), , drop = FALSE]
  )
}

significance_stars <- function(p) {
  ifelse(is.finite(p) & p < 0.001, "****",
    ifelse(is.finite(p) & p < 0.01, "***",
      ifelse(is.finite(p) & p < 0.05, "*", "")))
}

format_estimate <- function(estimate, p.value = NA_real_) {
  ifelse(is.finite(estimate), paste0(sprintf("%.3f", estimate),
    ↪ significance_stars(p.value)), NA_character_)
}

format_se <- function(std.error) {
  ifelse(is.finite(std.error), sprintf("%.3f", std.error), NA_character_)
}

regression_display_table <- function(terms, estimates, std_errors, p_values, outcome_label,
  ↪ gof = NULL) {
  rows <- safe_bind_rows(lapply(seq_along(terms), function(i) {
    data.frame(
      Term = c(terms[[i]], ""),
      value = c(format_estimate(estimates[[i]], p_values[[i]]), format_se(std_errors[[i]])),
      stringsAsFactors = FALSE
    )
  })))
  names(rows)[names(rows) == "value"] <- outcome_label
  if (!is.null(gof) && nrow(gof)) {
    names(gof) <- names(rows)
    rows <- rbind(rows, gof)
  }
  rows
}

first_finite_scalar <- function(value) {
  value <- suppressWarnings(as.numeric(value))
  value <- value[is.finite(value)]
  if (length(value)) value[[1]] else NA_real_
}

format_gof_number <- function(value, digits = 3L, integer = FALSE) {
  value <- first_finite_scalar(value)
  if (!is.finite(value)) return("")
  if (isTRUE(integer)) sprintf("%.0f", value) else sprintf(paste0("%.", digits, "f"), value)
}

```

```

}

model_gof_rows <- function(model, outcome_label) {
  if (is.null(model) || is_model_status_payload(model)) return(data.frame())
  sm <- tryCatch(summary(model), error = function(e) NULL)
  nobs <- tryCatch(stats::nobs(model), error = function(e) NA_real_)
  r2 <- tryCatch(sm$r.squared, error = function(e) NA_real_)
  adj_r2 <- tryCatch(sm$adj.r.squared, error = function(e) NA_real_)
  fstat <- tryCatch(sm$fstatistic[[1]], error = function(e) NA_real_)
  data.frame(
    Term = c("Observations", "R-squared", "Adjusted R-squared", "Model's F-Statistic"),
    value = c(
      format_gof_number(nobs, integer = TRUE),
      format_gof_number(r2),
      format_gof_number(adj_r2),
      format_gof_number(fstat, digits = 2L)
    ),
    check.names = FALSE,
    stringsAsFactors = FALSE
  ) |>
  stats::setNames(c("Term", outcome_label))
}

probit_gof_rows <- function(selection_model, n, outcome_label) {
  rows <- data.frame(
    Term = "Observations",
    value = format_gof_number(n, integer = TRUE),
    stringsAsFactors = FALSE
  )

  # The active participation model is fit with survey::svyglm(), which is
  # design-based rather than maximum-likelihood. Likelihood-based quantities
  # such as log-likelihood and McFadden's pseudo-R^2 are therefore deliberately
  # omitted for svyglm objects; calling stats::logLik() on svyglm emits the
  # survey package warning that the model was not fitted by maximum likelihood.
  # If this helper is reused for an ordinary glm in diagnostics, the usual ML
  # summary rows are still meaningful and are reported.
  if (inherits(selection_model, "glm") && !inherits(selection_model, "svyglm")) {
    loglik <- tryCatch(stats::logLik(selection_model), error = function(e) NA_real_)
    null_dev <- tryCatch(selection_model$null.deviance, error = function(e) NA_real_)
    dev <- tryCatch(selection_model$deviance, error = function(e) NA_real_)
    pseudo_r2 <- if (is.finite(first_finite_scalar(null_dev)) &&
      ↪ first_finite_scalar(null_dev) > 0 && is.finite(first_finite_scalar(dev))) {
      1 - first_finite_scalar(dev) / first_finite_scalar(null_dev)
    } else {
      NA_real_
    }
    rows <- rbind(
      rows,
      data.frame(Term = "Log Likelihood", value = format_gof_number(loglik, digits = 2L),
        ↪ stringsAsFactors = FALSE),
      data.frame(Term = "McFadden pseudo-R-squared", value = format_gof_number(pseudo_r2),
        ↪ stringsAsFactors = FALSE)
    )
  }
}

```

```

  stats::setNames(rows, c("Term", outcome_label))
}

numeric_summary <- function(df, variables = NULL) {
  df <- as.data.frame(df)
  if (is.null(variables)) variables <- names(df)
  variables <- intersect(variables, names(df))
  out <- lapply(variables, function(v) {
    x <- suppressWarnings(as.numeric(as.character(df[[v]])))
    x <- x[is.finite(x)]
    if (!length(x)) return(NULL)
    data.frame(variable = v, min = min(x), q1 = unname(stats::quantile(x, 0.25)), median =
      ↪ stats::median(x), q3 = unname(stats::quantile(x, 0.75)), max = max(x), mean =
      ↪ mean(x), sd = stats::sd(x), n = length(x), stringsAsFactors = FALSE)
  })
  out <- safe_bind_rows(out)
  if (!nrow(out)) data.frame(variable = character(), n = integer()) else out
}

categorical_summary <- function(df, variables) {
  df <- as.data.frame(df)
  variables <- intersect(variables, names(df))
  safe_bind_rows(lapply(variables, function(v) {
    x <- df[[v]]
    tab <- sort(table(x, useNA = "ifany"), decreasing = TRUE)
    if (!length(tab)) return(NULL)
    data.frame(variable = v, category = names(tab), n = as.integer(tab), share =
      ↪ round(as.numeric(tab) / sum(tab), 4), stringsAsFactors = FALSE)
  )))
}

table_status_row <- function(model, status = "unavailable", reason = NA_character_) {
  data.frame(
    model = model,
    term = NA_character_,
    estimate = NA_real_,
    std.error = NA_real_,
    statistic = NA_real_,
    p.value = NA_real_,
    status = status,
    reason = reason,
    stringsAsFactors = FALSE
  )
}

table_first_existing_column <- function(df, candidates) {
  hit <- intersect(candidates, names(df))
  if (length(hit)) hit[[1]] else NA_character_
}

table_column_or_na <- function(df, column) {
  if (length(column) != 1L || is.na(column) || !column %in% names(df)) {
    return(rep(NA_real_, nrow(df)))
  }
}

```

```

  suppressWarnings(as.numeric(df[[column]]))
}

coefficient_frame <- function(model, vcov_matrix = NULL) {
  estimates <- tryCatch(stats::coef(model), error = function(e) NULL)
  if (is.null(estimates) || !length(estimates)) return(data.frame())

  terms <- names(estimates)
  if (is.null(terms) || !length(terms)) terms <- paste0("term_", seq_along(estimates))
  estimates <- suppressWarnings(as.numeric(estimates))

  vc <- vcov_matrix
  if (is.null(vc)) vc <- tryCatch(stats::vcov(model), error = function(e) NULL)
  se <- rep(NA_real_, length(estimates))
  if (!is.null(vc) && length(dim(vc)) == 2L && all(dim(vc) >= length(estimates))) {
    diag_vc <- suppressWarnings(as.numeric(diag(vc)))
    vc_terms <- rownames(vc)
    if (!is.null(vc_terms) && length(vc_terms)) {
      matched <- match(terms, vc_terms)
      ok <- !is.na(matched) & matched <= length(diag_vc)
      se[ok] <- sqrt(pmax(diag_vc[matched[ok]], 0))
    } else {
      se <- sqrt(pmax(diag_vc[seq_along(estimates)], 0))
    }
  }

  statistic <- estimates / se
  statistic[!is.finite(statistic)] <- NA_real_
  df_resid <- tryCatch(stats::df.residual(model), error = function(e) NA_real_)
  p_value <- if (is.finite(df_resid) && df_resid > 0) {
    2 * stats::pt(abs(statistic), df = df_resid, lower.tail = FALSE)
  } else {
    2 * stats::pnorm(abs(statistic), lower.tail = FALSE)
  }
  p_value[!is.finite(p_value)] <- NA_real_

  out <- data.frame(
    Estimate = estimates,
    `Std. Error` = se,
    statistic = statistic,
    `Pr(>|t|)` = p_value,
    check.names = FALSE,
    stringsAsFactors = FALSE
  )
  rownames(out) <- terms
  out
}

plain_model_coefficients <- function(model) {
  coefficient_frame(model)
}

clustered_model_coefficients <- function(model) {
  tryCatch({
    vc_fun <- clustered_model_vcov(model)

```

```

    if (is.null(vc_fun)) return(data.frame())
    vc <- vc_fun(model)
    coefficient_frame(model, vc)
  }, error = function(e) data.frame())
}

filter_table_model <- function(df, candidates) {
  if (!"model" %in% names(df)) return(df)
  out <- df[df$model %in% candidates, , drop = FALSE]
  if (nrow(out)) out else df[0L, , drop = FALSE]
}

is_model_status_payload <- function(x) {
  is.list(x) &&
  !inherits(x, c("lm", "ivreg")) &&
  !is.null(x$status) &&
  length(x$status) == 1L &&
  (is.character(x$status) || is.factor(x$status))
}

model_status_reason <- function(x) {
  reason <- x$reason
  if (is.null(reason) || !length(reason) || all(is.na(reason))) NA_character_ else
    ↪ as.character(reason[[1]])
}

tidy_iv_models <- function(iv_models) {
  if (!is.list(iv_models) || inherits(iv_models, c("lm", "ivreg"))) iv_models <- list(model
    ↪ = iv_models)
  safe_bind_rows(lapply(names(iv_models), function(name) {
    model <- iv_models[[name]]
    if (is_model_status_payload(model)) {
      return(table_status_row(name, as.character(model$status[[1]]),
        ↪ model_status_reason(model)))
    }
    coefs <- clustered_model_coefficients(model)
    if (!nrow(coefs)) coefs <- plain_model_coefficients(model)
    if (!nrow(coefs)) {
      return(table_status_row(name, "unavailable", "Model coefficients are unavailable."))
    }
    estimate_col <- table_first_existing_column(coefs, c("Estimate", "estimate"))
    se_col <- table_first_existing_column(coefs, c("Std. Error", "std.error", "Std.Error"))
    statistic_col <- table_first_existing_column(coefs, c("t value", "z value", "t", "z",
    ↪ "statistic"))
    p_col <- table_first_existing_column(coefs, c("Pr(>|t|)", "Pr(>|z|)", "p.value", "p",
    ↪ "P>|t|", "P>|z|"))

    if (is.na(estimate_col)) {
      return(table_status_row(name, "unavailable", paste("Model coefficient table lacks an
        ↪ estimate column. Columns:", paste(names(coefs), collapse = ", "))))
    }

    data.frame(
      model = name,

```

```

      term = rownames(coefs),
      estimate = table_column_or_na(coefs, estimate_col),
      std.error = table_column_or_na(coefs, se_col),
      statistic = table_column_or_na(coefs, statistic_col),
      p.value = table_column_or_na(coefs, p_col),
      status = "estimated",
      reason = NA_character_,
      stringsAsFactors = FALSE
    )
  }
})
}

clustered_model_vcov <- function(model) {
  cluster <- attr(model, "cluster_state")
  if (is.null(cluster)) return(NULL)
  cluster <- as.vector(cluster)
  if (length(cluster) != stats::nobs(model)) return(NULL)
  if (sum(!is.na(cluster)) < 2L || length(unique(cluster[!is.na(cluster)])) <= 1L)
    ↪ return(NULL)
  if (!requireNamespace("sandwich", quietly = TRUE)) return(NULL)
  force(cluster)
  function(x) sandwich::vcovCL(x, cluster = cluster)
}

#' make tables
#'
#' @return A named list of data frames consumed by save_tables().
make_tables <- function(selection_data, ame_results, district_panel, iv_models,
  ↪ first_stage_tests, cfg, selection_model = NULL) {
  cons_iv <- tidy_iv_models(iv_models)
  cons_iv_required <- filter_table_model(cons_iv, c("consumption", "baseline"))
  table_failures <- c(
    table_status_failures(ame_results, cfg, "average marginal effects"),
    table_status_failures(first_stage_tests, cfg, "first-stage regression"),
    table_status_failures(cons_iv_required, cfg, "second-stage IV regression")
  )

  fs_cons <- make_first_stage_table(first_stage_tests, cfg)
  cons_iv_table <- make_second_stage_table(iv_models)
  fs_model <- first_stage_table_model(iv_models, district_panel)
  if (!is.null(fs_model$model)) {
    attr(fs_cons, "table_model") <- fs_model$model
    attr(fs_cons, "table_vcov") <- fs_model$vcov
    attr(fs_cons, "table_add_rows") <- fs_model$add_rows
  }
  cons_model <- second_stage_table_model(iv_models)
  if (!is.null(cons_model$model)) {
    attr(cons_iv_table, "table_model") <- cons_model$model
    attr(cons_iv_table, "table_vcov") <- cons_model$vcov
  }

  out <- list(
    selection_n = data.frame(n = nrow(as.data.frame(selection_data))),
    sum_tbl_probit_quant = make_selection_summary_numeric_table(selection_data),
    sum_tbl_probit_cat = make_selection_summary_categorical_table(selection_data),

```

```

    probit_mfx = make_probit_ame_table(ame_results, nrow(as.data.frame(selection_data)),
    ↪ selection_model),
    sum_tbl_iv = make_iv_summary_table(district_panel),
    fs_cons = fs_cons,
    cons_iv = cons_iv_table,
    ame_results = as.data.frame(ame_results),
    first_stage = as.data.frame(first_stage_tests)
  )
  table_failures <- c(table_failures, attr(out$fs_cons, "table_input_failures"))
  table_failures <- unique(stats::na.omit(table_failures))
  if (length(table_failures)) attr(out, "table_input_failures") <- table_failures
  out
}

make_selection_summary_numeric_table <- function(selection_data) {
  meta <- data.frame(
    var = c("AGE", "HH_SIZE", "ENROLLMENT_COST", "dmean_num_IS_EDU_FREE",
    ↪ "dmean_num_TUTION_FEE_WAIVED", "dmean_num_RECD_SCHOLARSHIP_STIPEND",
    ↪ "dmean_num_RECD_TXT_BOOKS", "dmean_num_RECD_STATIONERY",
    ↪ "dmean_num_MID_DAY_MEAL_ETC_RECD", "dmean_num_ENROLLMENT_COST"),
    label = c("Age", "Household size", "Enrollment cost (Rs.)", "Educ. free available? (Yes
    ↪ = 1)", "Tuition waived?", "Scholarship/Stipend?", "Textbooks received?", "Stationery
    ↪ received?", "Mid-day meal or more received?", "Avg. district enrollment cost"),
    stringsAsFactors = FALSE
  )
  out <- public_numeric_stats(selection_data, meta, cost_vars = "ENROLLMENT_COST")
  insert_summary_group(out, "District-level aggregates:", "dmean_num_IS_EDU_FREE")
}

make_selection_summary_categorical_table <- function(selection_data) {
  meta <- data.frame(
    var = c("SEX", "RELIGION", "SOCIAL_GROUP", "SECTOR", "DIST_FROM_NEAREST_PRIMARY_CLASS",
    ↪ "father_educ"),
    label = c("Sex", "Religion", "Social group", "Urban", "Distance of nearest primary
    ↪ class", "Father's education"),
    stringsAsFactors = FALSE
  )
  public_categorical_stats(selection_data, meta)
}

make_probit_ame_table <- function(ame_results, n = NA_integer_, selection_model = NULL) {
  native_ame <- attr(ame_results, "marginaleffects_object", exact = TRUE)
  out <- as.data.frame(ame_results, stringsAsFactors = FALSE)
  if (!"Term" %in% names(out)) out$Term <- out$term
  table <- data.frame(
    Term = out$Term,
    Estimate = format_estimate(
      suppressWarnings(as.numeric(out$estimate)),
      suppressWarnings(as.numeric(out$p.value))
    ),
    `Std. Error` = format_se(suppressWarnings(as.numeric(out$std.error))),
    check.names = FALSE,
    stringsAsFactors = FALSE
  )
  if (!is.null(native_ame)) {

```



```

    attr(table, "marginaleffects_object") <- native_ame
    attr(table, "marginaleffects_n") <- n
  }
  if (!is.null(selection_model)) {
    attr(table, "selection_model") <- selection_model
  }
  table
}

make_iv_summary_table <- function(district_panel) {
  meta <- data.frame(
    var = c(
      "wavg_ling_degrees", "EMIE",
      "npeople_0708", "consumption_0708", "gini_cons_0708", "pct_urban", "avg_hh_size",
      "dependency_ratio", "pct_fem_head", "pct_hindu", "pct_muslim", "pct_other_religion",
      "pct_st", "pct_sc", "pct_obc", "pct_small_land", "pct_medium_land", "pct_large_land",
      "pct_head_illiterate", "pct_head_lit_to_primary", "pct_head_secondary_plus",
      ↪ "pct_pucca",
      "npeople_1718", "consumption_1718", "gini_cons_1718",
      "consumption_pct_change", "gini_change"
    ),
    label = c(
      "Ling. Distance",
      "EMIE", "Population", "Consumption", "Gini of Consumption", "Pct. Urban", "Avg. HH
      ↪ Size",
      "Dependency Ratio × 100", "Pct. Female Head", "Pct. Hindu", "Pct. Muslim", "Pct.
      ↪ Other",
      "Pct. ST", "Pct. SC", "Pct. OBC", "Pct. Small Land-Owner", "Pct. Med. Land-Owner",
      ↪ "Pct. Large Land-Owner",
      "Pct. Head Educ., Illiterate", "Pct. Head Educ., Lit.-Primary", "Pct. Head Educ.,
      ↪ Secondary+", "Pct. Pucca",
      "Population", "Consumption", "Gini of Consumption",
      "Percent change in consumption", "Change in Gini of consumption"
    ),
    desc = c(
      "Average linguistic distance of mother tongue from Hindi",
      "EMI exposure",
      "Estimated via NSS sample weights",
      "Average household monthly consumption expenditures (Rs.)",
      "Gini coefficient of consumption",
      "Percentage of people in an urban area",
      "Average household size",
      "Ratio of dependents (0-14, 65+) to labor force (15-64), × 100",
      "Percentage of households with a female head",
      "Percentage of Hindus",
      "Percentage of Muslims",
      "Percentage not Hindu/Muslim",
      "Scheduled Tribe",
      "Scheduled Caste",
      "Other Backward Class",
      "Owns 0.005-0.40 hectares",
      "Owns 0.41-3.00 hectares",
      "Owns $\\geq$ 3.01 hectares",
      "Percentage of household heads with educ. level: illiterate",
      "Percentage of heads with educ. level: literate-primary",

```

```

    "Percentage of heads with educ. level: above secondary",
    "Percentage in pucca (permanent) homes",
    "Estimated via NSS sample weights",
    "Average household monthly consumption expenditures (Rs.)",
    "Gini coefficient of consumption",
    "Percent change in consumption",
    "Change in the Gini coefficient of consumption"
  ),
  stringsAsFactors = FALSE
)
out <- public_numeric_stats(district_panel, meta, count_vars = c("npeople_0708",
↪ "npeople_1718"))
out <- insert_summary_group(out, "From 2001:", "wavg_ling_degrees")
out <- insert_summary_group(out, "From 2007-08:", "EMIE")
out <- insert_summary_group(out, "From 2017-18:", "npeople_1718")
out <- insert_summary_group(out, "From 2007-08 to 2017-18:", "consumption_pct_change")
out
}
make_first_stage_table <- function(first_stage_tests, cfg = list()) {
  fs <- as.data.frame(first_stage_tests, stringsAsFactors = FALSE)
  required_cols <- c("model", "term", "estimate", "std.error", "p.value", "partial_f",
↪ "partial_p", "status")
  missing_cols <- setdiff(required_cols, names(fs))
  if (length(missing_cols)) {
    msg <- paste("First-stage results are missing columns:", paste(missing_cols, collapse =
↪ ", "))
    if (is_final_mode(cfg)) stop(msg, call. = FALSE)
    return(table_status_row("first_stage", "unavailable", msg))
  }
  if ("model" %in% names(fs) && any(fs$model %in% c("consumption", "baseline"))) fs <-
↪ fs[fs$model %in% c("consumption", "baseline"), , drop = FALSE]
  if (any(grepl("[0-9]+$", as.character(fs$term)))) {
    msg <- "First-stage coefficient terms are numeric row positions; coefficient names were
↪ lost upstream."
    if (is_final_mode(cfg)) stop(msg, call. = FALSE)
    return(table_status_row("first_stage", "malformed", msg))
  }
  status_reasons <- unique(stats::na.omit(as.character(fs$reason[!is.na(fs$status) &
↪ fs$status != "estimated"])))
  fs <- fs[fs$status == "estimated" & !is.na(fs$term), , drop = FALSE]
  required_terms <- c("wavg_ling_degrees", "(Intercept)")
  missing_terms <- setdiff(required_terms, fs$term)
  if (length(missing_terms)) {
    if (length(status_reasons)) {
      msg <- paste(status_reasons, collapse = "; ")
      out <- table_status_row("first_stage", "out_of_active_pipeline", msg)
      attr(out, "table_input_failures") <- paste0("First-stage table lacks required
↪ coefficient(s): ", paste(missing_terms, collapse = ", "), ". Reasons: ", msg)
      return(out)
    }
    msg <- paste("First-stage table lacks required coefficient(s):", paste(missing_terms,
↪ collapse = ", "))
    if (is_final_mode(cfg)) stop(msg, call. = FALSE)
    return(table_status_row("first_stage", "unavailable", msg))
  }
}

```

```

term_order <- c(
  "wavg_ling_degrees", "consumption_0708", "gini_cons_0708",
  "pct_urban", "avg_hh_size", "dependency_ratio", "pct_fem_head",
  "pct_hindu", "pct_muslim", "pct_st", "pct_sc", "pct_obc",
  "pct_small_land", "pct_medium_land", "pct_large_land",
  "pct_head_lit_to_primary", "pct_head_secondary_plus", "(Intercept)"
)
fs <- fs[order(match(fs$term, term_order), fs$term), , drop = FALSE]
fs <- fs[!is.na(match(fs$term, term_order)), , drop = FALSE]

stat <- fs[fs$term == "wavg_ling_degrees", , drop = FALSE]
f_value <- if (nrow(stat)) first_finite_value(stat, c("partial_f", "model_f")) else
  NA_real_
f_p <- if (nrow(stat)) first_finite_value(stat, c("partial_p", "model_p")) else NA_real_
f_row <- data.frame(
  Term = "Instrument's F-Statistic",
  value = paste0(sprintf("%.2f", f_value), significance_stars(f_p)),
  stringsAsFactors = FALSE
)
nobs_value <- first_finite_value(stat, c("nobs", "n", "N"))
r2_value <- first_finite_value(stat, c("r.squared", "r2"))
adj_r2_value <- first_finite_value(stat, c("adj.r.squared", "adj_r2"))
gof <- safe_bind_rows(list(
  data.frame(Term = "Observations", value = ifelse(is.finite(nobs_value), sprintf("%.0f",
    nobs_value), ""), stringsAsFactors = FALSE),
  data.frame(Term = "R-squared", value = ifelse(is.finite(r2_value), sprintf("%.3f",
    r2_value), ""), stringsAsFactors = FALSE),
  data.frame(Term = "Adjusted R-squared", value = ifelse(is.finite(adj_r2_value),
    sprintf("%.3f", adj_r2_value), ""), stringsAsFactors = FALSE),
  f_row
))

regression_display_table(
  terms = iv_table_term_label(fs$term),
  estimates = suppressWarnings(as.numeric(fs$estimate)),
  std_errors = suppressWarnings(as.numeric(fs$std.error)),
  p_values = suppressWarnings(as.numeric(fs$p.value)),
  outcome_label = "EMI Exposure",
  gof = gof
)
}

first_finite_value <- function(df, cols) {
  for (col in cols) {
    if (!col %in% names(df)) next
    value <- suppressWarnings(as.numeric(df[[col]][[1]]))
    if (length(value) && is.finite(value)) return(value)
  }
  NA_real_
}

first_stage_table_model <- function(iv_models, district_panel) {
  model <- second_stage_table_model(iv_models)$model
  if (is.null(model) || !inherits(model, "ivreg")) return(list(model = NULL, vcov = NULL,
    add_rows = NULL))

```

```

terms <- parse_iv_formula_terms(model)
if (is.null(terms) || !length(terms$regressors) || !length(terms$instruments)) {
  return(list(model = NULL, vcov = NULL, add_rows = NULL))
}
endogenous <- setdiff(terms$regressors, terms$instruments)
if (!length(endogenous)) endogenous <- terms$regressors[[1]]
if (!length(endogenous) || is.na(endogenous[[1]]) || !nzchar(endogenous[[1]])) {
  return(list(model = NULL, vcov = NULL, add_rows = NULL))
}
formula <- stats::as.formula(paste(endogenous[[1]], "~", paste(terms$instruments, collapse
↪ = " + ")))
data <- add_iv_panel_aliases(district_panel)
if (length(setdiff(all.vars(formula), names(as.data.frame(data)))) {
  return(list(model = NULL, vcov = NULL, add_rows = NULL))
}
fit <- tryCatch(stats::lm(formula, data = data), error = function(e) NULL)
if (is.null(fit)) return(list(model = NULL, vcov = NULL, add_rows = NULL))
vc <- first_stage_vcov(fit, data)
excluded <- setdiff(terms$instruments, terms$regressors)
excluded_term <- if (length(excluded)) excluded[[1]] else NA_character_
wald <- first_stage_wald_test(fit, excluded_term, vc)
add_rows <- data.frame(
  term = "Instrument's F-Statistic",
  estimate = paste0(formatC(wald$partial_f, digits = 2, format = "f"),
  ↪ significance_stars(wald$partial_p)),
  stringsAsFactors = FALSE
)
list(model = fit, vcov = vc, add_rows = add_rows)
}

second_stage_table_model <- function(iv_models) {
  model <- NULL
  if (is.list(iv_models) && !inherits(iv_models, c("lm", "ivreg"))) {
    hit <- intersect(c("consumption", "baseline"), names(iv_models))
    if (length(hit)) model <- iv_models[[hit[[1]]]]
  } else {
    model <- iv_models
  }
  if (is.null(model) || is_model_status_payload(model)) return(list(model = NULL, vcov =
  ↪ NULL))
  vc_fun <- clustered_model_vcov(model)
  vc <- if (is.null(vc_fun)) NULL else tryCatch(vc_fun(model), error = function(e) NULL)
  list(model = model, vcov = vc)
}

make_second_stage_table <- function(iv_models) {
  out <- tidy_iv_models(iv_models)
  out <- filter_table_model(out, c("consumption", "baseline"))
  if (!nrow(out)) return(data.frame(Term = character(), `Consumption Growth` = character(),
  ↪ check.names = FALSE))
  model <- NULL
  if (is.list(iv_models) && !inherits(iv_models, c("lm", "ivreg"))) {
    hit <- intersect(c("consumption", "baseline"), names(iv_models))
    if (length(hit)) model <- iv_models[[hit[[1]]]]
  } else {

```

```

    model <- iv_models
  }
  regression_display_table(
    terms = iv_table_term_label(out$term),
    estimates = suppressWarnings(as.numeric(out$estimate)),
    std_errors = suppressWarnings(as.numeric(out$std.error)),
    p_values = suppressWarnings(as.numeric(out$p.value)),
    outcome_label = "Consumption Growth",
    gof = model_gof_rows(model, "Consumption Growth")
  )
}

iv_table_term_label <- function(term) {
  labels <- c(
    "(Intercept)" = "Constant",
    EMIE = "EMIE",
    emie_2007 = "EMIE",
    wavg_ling_degrees = "Linguistic distance",
    consumption_0708 = "Consumption, 2007-08",
    gini_cons_0708 = "Gini consumption, 2007-08",
    pct_urban = "Pct. urban",
    avg_hh_size = "Avg. HH size",
    dependency_ratio = "Dependency ratio",
    pct_fem_head = "Pct. female head",
    pct_hindu = "Pct. Hindu",
    pct_muslim = "Pct. Muslim",
    pct_st = "Pct. ST",
    pct_sc = "Pct. SC",
    pct_obc = "Pct. OBC",
    pct_small_land = "Pct. small land",
    pct_medium_land = "Pct. medium land",
    pct_large_land = "Pct. large land",
    pct_head_lit_to_primary = "Pct. head literate-primary",
    pct_head_secondary_plus = "Pct. head secondary+"
  )
  out <- unname(labels[term])
  out[is.na(out)] <- term[is.na(out)]
  out
}

```